

Mixture of Limits: Navigation Law for Computer Intelligence

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Research study · OpenIE research.openie.dev

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Measurement. Research study, not a journal final. Energy figures from the software reference are OpCounter analytical estimates, not board power. We have not synthesized or metered an FPGA board. No fabricated citations.

Floors where more bits stop buying outcomes. Compression and predictive formulas beat excess enumeration; the model leaf is a conditionally available residual, off by default, never coercible to Deterministic. Soft-ref means the software reference path; not FPGA or board package energy.

Abstract

Mixture of Limits is a navigation law for computer intelligence: there exist **floors** past which additional tokens, parameters, or joules do not purchase verifiable progress on a task coordinate. The law is information-theoretic and rooted in physics. Across human scientific history, compression into predictive formulas and invariants has repeatedly beaten excess enumeration of observations: from Kepler's laws over Tycho's tables, through Newton's closed forms, to Shannon's bit accounting and Landauer's thermodynamic floor on irreversible erasure. Mixture of Limits operationalizes that lineage for machines: **Lookup** → **Formula** → **Solver/settle** → **Model LAST**, with close owned as propose → certify → commit/refuse → receipt. Unstructured front doors (speech, pixels, free text) need a **dual-phase** stack: Phase 1 is a bounded ultra-light quantized transducer that emits a typed AST or schema; Phase 2 is the Mixture of Limits cascade. Perception is a front gear, not a demotion of Model LAST for residual reasoning.

The industry default escalates Mixture-of-Experts (MoE) capacity *inside* a generative corridor. Mixture of Limits instead names floors *outside* generation: Value of Information (Vol), grammar coverage, Landauer/joule estimate, certificate, and settle-refuse. The industry race is a **race to those plateau floors**, not unbounded scale: once Vol is zero past the floor, optimize energy/compute on the plateau. Mixture of Limits demotes the neural net to a residual leaf. Soft-ref mol prove is constructive existence in software. Catalog and Landauer figures on receipts are **estimates**; package measured_j appears only when a labeled meter returns a reading.

Terms used here

Soft-ref has one meaning in this paper: the software reference path that proves the navigation law without board synthesis or package metering. Receipts on that path label energy as estimate and leave `measured_j` unset until a named meter returns. Soft-ref is not a second, looser evidence class for live silicon. Periodic Stack, replay class, value of information, evidence classes, and `board_synth_claimed` are defined on the shared [glossary](#).

Model leaf (one sentence). The model gear is a conditionally available residual: off by default (`allow_model=false`), opened only when cheaper gears miss, and never coercible to Deterministic (ModelGenerated cannot be rewritten as Deterministic).

Living figures for claims on this page. Cascade: [/living/mol/#mol-diag-01](#). Named floors: [/living/mol/#mol-diag-02](#). Commit or refuse then receipt: [/living/mol/#mol-diag-03](#). MoE contrast: [/living/mol/#mol-tab-01](#).

The law before the survey

Lookup → Formula → Solver → Model last. Floors stop escalation when more bits stop buying outcomes: Vol, grammar, energy estimate, certificate, settle-refuse. Close is propose → certify → commit/refuse → receipt. Section 8 is a survey of plateau options under that law. It is not the law.

Each class of embodiment reaches a limit. A decision model stops at a typed probability. A bounded ask stops at a typed value or at not-sure. A statechart stops at a legal transition. An energy-based settle stops at a low-energy configuration. A visual editor stops at the chart a person can audit. None of them is the whole intelligence. The convergence is hybrid interplay: the chart is the door, the decision model may fill a proposal, the energy settle may score a configuration, and something outside the model still commits or refuses. Mixture of Limits is the navigation law of that interplay. Refuse is success. Energy to run is the only true metric. All other factors collapse to zero. §8.5.1 places the public instances under that law. It does not replace the law.

1. Introduction: pursuit of limits toward AGI

The dominant construction for computer intelligence (CI) treats the neural net as the substrate and excess tokens as the growth variable. That construction diverges from the historical method that produced reliable science. When Brahe enumerated planetary positions, the tables were necessary but not the law. Kepler compressed years of positions into three predictive relations. Newton further compressed those relations into invariants and closed forms that travel across domains. The same pattern recurs wherever a grammar of nature is covered: a short formula outruns an infinite ledger of cases.

Information theory names the modern accounting of that pattern. Shannon (1948) prices bits under uncertainty as settled law. Value of Information (Vol) prices whether another observation is worth its cost for a decision. Landauer (1961) prices irreversible bit erasure in joules at temperature T : $E_{\min}$

$= k_B T \ln 2$ per bit erased in the ideal model. That Landauer quantity is a thermodynamic lower bound and, on Mixture of Limits receipts, a **labeled estimate**—not a wattmeter reading. Together they imply floors: past a point, more bits stop buying outcomes that matter for a stated benefit.

Thesis. Pursuit of limits is the path toward AGI-grade reliability. Excess-token and MoE scaling diverge from true Vol for a given benefit. Mixture of Limits is the executable navigation law that binds those floors (Vol, grammar, Landauer/joules as estimate, certificate, settle-refuse) against industry Mixture-of-Experts. **The industry race is a race to the plateau floors, not unbounded scale:** once Vol is zero past a named floor, more parameters and tokens buy nothing verifiable on that coordinate—only energy/compute optimization on the plateau remains. **Addendum:** Shannon, Landauer, Kolmogorov/Solomonoff/Chaitin, Howard (Vol), and formula-first science AI already state these floors as settled theory. The remaining bottleneck is **applied mathematics embodied in materials and physical hardware. Frontier addendum (§8.4-§8.14): SOTA at the plateau.** Once the floor is known and Value of Information is zero past it, the only remaining work is optimize energy and compute on that plateau: cheapest-sufficient Lookup → Formula → Solver → Model LAST. Deeper latents, energy-based hybrids, world models, test-time compute, super learning, SSM hybrids, mixture of experts, transmission / gearing / cascade / memory-context / test-time methods, and the cross-field plateau moves in §8.14 are **implementation options for plateau spend**—not a narrative race among generators. They do not cancel Vol, grammar, energy, or certify floors. Mixture of Limits remains the navigation law. §8.5.1 is a separate claim, not another plateau option: each embodiment class stops at its own object, and the convergence is hybrid interplay under this law. Dual-phase perception (§11.1) bounds the unstructured front door without retiring Model LAST for residual reasoning. The hardware-economics reading (§8.12) states that computers are hardware, software is applied engineering under constraints, and plateau spend still binds to floors on real devices. §8.13 organizes verified public reports by technique—transmission, gearing/MoE routing, cascade, memory/context, and test-time—as sources for plateau options. §8.14 races major AI/ML fields to the floor that binds each.

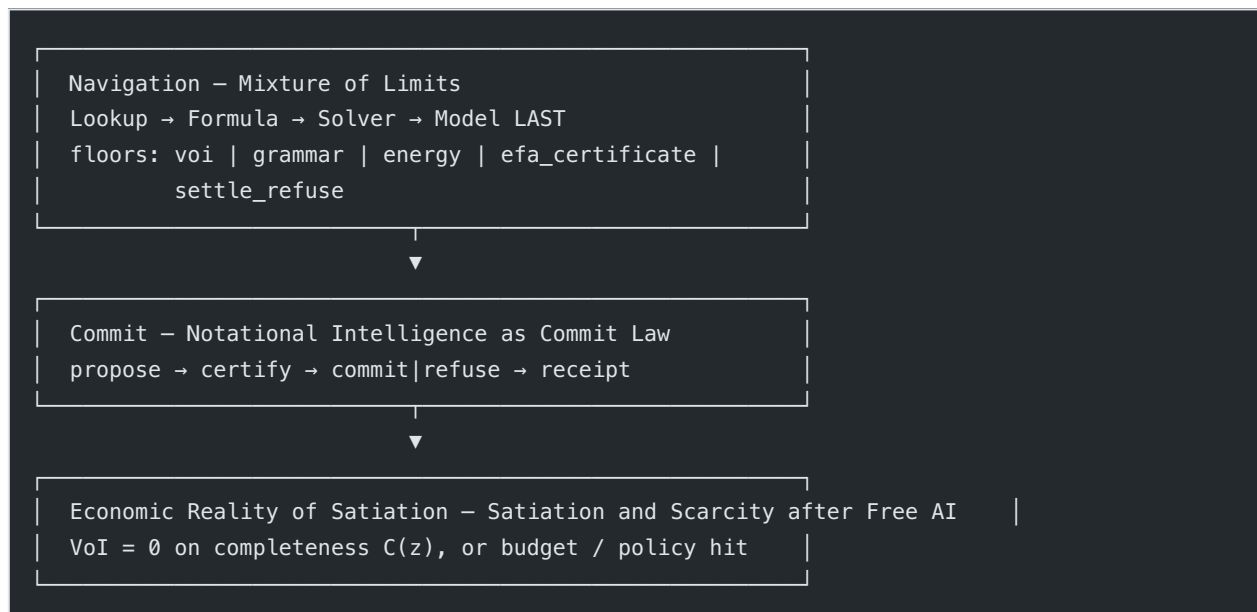
This paper's contribution is the study prose for that law as published on research.openie.dev, grounded in the clean-room `mixture-of-limits` software reference (Apache-2.0 OR MIT). Companion studies already on this site supply the commit-record interface ([Notational Intelligence as Commit Law](#)) and the economic stop after free digital inference ([Satiation and Scarcity after Free AI](#)). Mixture of Limits is the navigation law those companions sit under: NI owns irreversible commit shape; Satiation owns Economic Reality of Satiation; Mixture of Limits owns *which gear closes and when refuse is success*.

Three measurement facts constrain every later number. First, joules from catalog surrogates and OpCounter-style analytics are **estimates**, not board power. Second, Landauer annotations are **estimates**, distinct from `RAPL/NVML/measured_j`. Third, `board_synth_claimed=false`; package `measured_j` is set only when a Metered probe returns a reading.

Companion laws

Three studies on this catalog form one stack: **Navigation**, **Commit**, and **Economic Reality of Satiation**. They compose. They do not collapse into one paper. A fourth study, [Metabolic Intelligence](#), owns the energy-budget envelope and actuators that make those floors bind at tag and campus scale.

Role	Study	Owns
Navigation	Mixture of Limits	Which gear closes: Lookup → Formula → Solver → Model LAST. Floors include Value of Information (VoI), grammar coverage, energy estimate, Energy-First Architecture (EFA) / certificate refuse, and settle-refuse.
Commit	Notational Intelligence as Commit Law	Irreversible close shape: propose → certify → commit refuse → receipt. Notational Intelligence owns the commit record. Analytical joules here are estimates; package measured_j only when Metered.
Economic Reality of Satiation	Satiation and Scarcity after Free AI	Stop when VoI is zero on a stated completeness predicate, or when budget / policy refuse fires. Free at the margin for digital inference is not free joules and not free actuation.
Energy budget / metabolic embodiment	Metabolic Intelligence	Budget envelope and actuators: digital enzymes, $J(m \mid q)$ obtain-router, OpenADR/MQTT campus service. Soft-ref path: board_synth_claimed=false.
Automation path	Spell Check is Global	Existence proof: computer intelligence that became ordinary because it was cheap, local, and paired to hardware people already have. Access framing, not a census: the many (7B+), not the few who rent frontier datacenters (<500M).



```

flowchart TB
  NAV["Navigation<br/>Mixture of Limits<br/>Lookup → Formula → Solver → Model LAST"]
  COM["Commit<br/>Notational Intelligence as Commit Law<br/>propose → certify → commit|  
refuse → receipt"]
  ECO["Economic Reality of Satiation<br/>Satiation and Scarcity after Free AI<br/>VoI = 0 or  
budget hit"]
  NAV --> COM --> ECO

```

Mixture of Limits owns *which gear closes* and *when refuse is success*. Notational Intelligence owns the irreversible commit shape. Satiation owns Economic Reality of Satiation. [Spell Check is Global](#) is the existence proof of the path: the nonword gear is Lookup, then a short edit, on hardware that is available, accessible, and capable. Soft-ref path: `board_synth_claimed=false`; `estimates ≠ measured_j`. Energy to run is the only true metric of computer intelligence: token price, parameter count, moat rent, and access fees collapse to zero by commoditization, and what remains is joules. Estimates are not `measured_j`.

OpenIE map (no prior literacy assumed)

This study lives inside the OpenIE family of sites. Readers do not need those sites memorized; the map below states what each surface is and where Mixture of Limits sits.

Surface	URL	What it is	Relation to Mixture of Limits
Research (this hub)	research.openie.dev	Readable studies, PDFs, and living figures. This paper is <code>/papers/mol/</code> ; companions are Notational Intelligence (<code>/papers/ni/</code>), Satiation (<code>/papers/satiation/</code>), Metabolic Intelligence (<code>/papers/mei/</code>), and Spell Check is Global (<code>/papers/spellcheck/</code>).	Publishes the navigation-law study prose and soft-ref measurement bounds.
Stack	stack.openie.dev	Teaching map of the family: information theory, game theory, and mechanism design as one substrate; directory of the eight periodic stacks.	Orientation layer. Mixture of Limits is not “another stack card”; it is the navigation law that chooses cheapest-sufficient close across stack coordinates.
Compute	compute.openie.dev	Periodic Stack of Computation: 258 primitives / 33 families , thermodynamic floor every sibling inherits.	The primitive table Mixture of Limits navigates . Soft-ref proves the full live catalog (258 Present / 258 live Lookup/Formula/Solver/Navigate gears) plus honest HW Gap cells; HW Gaps soft-ref sims x8 (<code>physical_settle...photonic_mzi</code>) +

Surface	URL	What it is	Relation to Mixture of Limits
			Ferric/Stage C inventories are in proof with Gap cells retained. μ calib + silicon meters remain outside soft-ref proof. Empty HW cells $\rightarrow$ primitive_gap. estimates $\neq$ measured_j; stage_c_measured=false until meters.
Knowledge	knowledge.openie.dev	Working definition of a claim as seven axes (valid time, transaction time, reference time, granularity, scope, certainty, provenance).	Typed claims and cite/compose leaves (Z2 cite / Z1 compose in soft-ref) bind to this object shape. Mixture of Limits does not redefine knowledge; it refuses escalation when grammar and Vol say the claim coordinate is already covered.
Synthesis	synthesis.openie.dev	Periodic Stack of Digital Information Synthesis: AI as software; zones $Z_1/Z_2/Z_3$; cost surface $E(x) = \sum \theta(p) \cdot \mu(p, H)$.	Plateau spend and cascade order rhyme with synthesis zones (closed-form $\rightarrow$ constrained $\rightarrow$ unbounded). Mixture of Limits owns floors outside generation —Vol, grammar, energy estimate, certify, settle-refuse—so Z_3 -style generation stays Model LAST.
Verify	proof.openie.dev	Periodic Stack of Verification: cheapest-sufficient solver under $E(x) \geq \theta(D) \cdot \mu(S, V)$.	Cousin energy law. Mixture of Limits uses the same spine form for path energy; Verify maps verified artifacts; Mixture of Limits maps which gear closes before commit.
Product	klere.ai	Klere — OpenIE product home. Metabolic Intelligence is designed for Klere (research owns the law/class; Klere owns product embodiment). Public site is thin access / energy-efficient AI framing (Oct 2026)—not a shipped feature catalog.	MEI envelope and Mixture of Limits navigation feed Klere's product embodiment. Soft-ref DX remains mol.yaml / mol run until Klere packaging ships.

One paragraph. Stack teaches the family map. Compute names the permitted-act primitives. Knowledge defines the claim object those primitives carry. Synthesis places digital information synthesis on a cost surface and zone grammar. Verify prices verified artifacts under a related energy inequality. Research hosts the studies. Klere (klere.ai) is the product home (MEI designed for Klere; thin public site—no invented features). **Mixture of Limits is the navigation law across that family:** Lookup $\rightarrow$ Formula $\rightarrow$ Solver $\rightarrow$ Model LAST, with named floors that stop escalation when more bits stop buying outcomes. It is not a rebrand of Mixture-of-Experts, not a substitute for the Periodic Stack pages, and not a claim that soft-ref software measures board package joules.

Other names this paper uses without treating them as assumed literacy: **MathGround** denotes the OpenIE cascade / replay-class discipline that enforces Lookup → Formula → Solver → Model LAST; **WCA** denotes capability / certify refuse adapters (live MCP path remains soft-ref roadmap); **leapfrog / openie-path** denotes ask-bridge adapters (stubs on the proven path). Living companions for this study: </living/mol/>. FPGA commit figures for NI: </living/fpga-sim/>.

Scope. Companion laws (Navigation / Commit / Economic Reality of Satiation) and the OpenIE map above name the catalog triad and stack / compute / knowledge / synthesis / verify / research before any later allusion. Section 2 states the law and Periodic Stack navigation. Section 3 states the proof spine $E(x) \geq \theta(D) \cdot \mu(S, V)$. Section 4 describes the cascade and close/receipt bind. Section 5 bridges to Satiation without rewriting it. Section 6 maps prove↔claim. Section 7 sketches Vol, grammar, and settle-refuse mathematics. Section 8 places related work as a verified citation chain (historical→recent proof points) plus Tier A formula/mechanism systems, then teaches latent space, Logical Intelligence (energy-based model / large language model / latent hybrid), the hybrid-interplay convergence of embodiment-class limits (§8.5.1), World Labs (spatial world models), and **SOTA at the plateau** (§8.7): once Vol is zero past the floor, plateau spend is the only work—test-time compute, super learning, live SOTA methods (SSMs, MoE, RAG, speculative decode, ...), SSM hybrids / MoE / implementation efficiency (including Jamba-class, DeepSeek, GLM as options), hardware economics (including edge / neuromorphic soft-ref), plateau techniques for transmission / gearing / cascade / memory-context / test-time (§8.13), industry gap levers (Mixture-of-Depths, AWQ/GPTQ/FP8, vLLM/SGLang; §8.10.1), and the cross-field **race to plateau floors** table (§8.14: RLHF, diffusion, GNNs, multimodal, federated/continual, pruning/KD/NAS, Bayesian/conformal/causal/active, GraphRAG, agent memory, structured generation, CUDA Graphs). Section 9 contrasts Mixture of Limits with Mixture-of-Experts and situates model-to-model field systems. Section 10 opens toward AGI via limits. Section 11 states engineering gaps to close—dual-phase front-door perception, O(1) meta-routing, Primitive Distillation for open grammars, Tier 0/1/2 measurement realism, and a declarative DX roadmap—as plateau work under the same floors, not apologies.

2. Law: floors where more bits stop buying outcomes

Mixture of Limits asserts: for a task coordinate, there exist floors such that additional tokens, parameters, or joules do not purchase verifiable progress. Intelligence is therefore a **navigation law** over a structured stack of primitives; not an invitation to dump residual capacity into a generative corridor.

Consequences:

1. **Mathematical compression / formulas beat excess tokens** when the grammar is covered.
2. **Neural nets are a conditionally available residual leaf** (off by default; never coercible to Deterministic), not the default substrate of CI.
3. Routing is **cheapest-sufficient**: escalate only on miss; refuse when escalation is unsafe or Vol-negative.

4. Proof / energy law: $E(x) \geq \theta(D) \cdot \mu(S, V)$ with labeled joule receipts; Landauer labeled as estimate.
5. **Available devices / multi-fabric**: after a cascade gear closes, pick the cheapest sufficient device class; not one accelerator by default.

Named floors

Limit id	Kind	Meaning
voi	ValueOfInformation	Marginal bits worthless under budget. stop
grammar	GrammarCoverage	Covered by Lookup/Formula/Solver; or refuse unknown
energy	Energy	Joule budget floor; Landauer-aware estimates
efa_certificate	Certificate	Diverge / fail certify → refuse before commit
settle_refuse	SettleRefuse	Ternary / energy-landscape settle will not settle → refuse
primitive_gap	PrimitiveGap	Missing Periodic Stack primitive; surface the gap
safety / wca_refuse	Safety / WCA	Capability deny; WCA certify refuse
information / latency	Information / Latency	Sufficiency and soft wall-clock ceilings

Neural net demotion

The model gear is a conditionally available residual: off by default (`allow_model=false`), opened only when cheaper gears miss, and never coercible to Deterministic (`ModelGenerated` cannot be rewritten as `Deterministic`). It may propose residual content; it never bypasses certify-before-commit. Coin-cell / edge power is $O(1)$ formula plus LUT refuse, not a tiny transformer.

Periodic Stack

Mixture of Limits navigates the OpenIE **Periodic Stack of Computation** at compute.openie.dev (**258 primitives / 33 families**). Sibling maps: stack.openie.dev (family directory), knowledge.openie.dev (seven-axis claim), synthesis.openie.dev (synthesis zones / cost surface). The soft-ref proves an in-tree **live catalog** navigator with 258 Present cells (all live Lookup/Formula/Solver/Navigate) + honest Gap markers and scale notes citing 258/33. HW Gaps soft-ref sims $\times 8$ + Ferric/Stage C inventories are in soft-ref proof (Gap cells retained; `stage_c_measured=false`). μ calibration corpora and silicon meters remain outside soft-ref proof scope. Empty cells surface as `primitive_gap`.

Mixture of Limits is not MoE

MoE (generative corridor)	Mixture of Limits (navigation law)
Experts live <i>inside</i> a model; gate selects parameters	Limits live <i>outside</i> generation; floors stop escalation
More experts → more capacity for the same generative act	More named floors → sharper refuse / cheaper close

MoE (generative corridor)	Mixture of Limits (navigation law)
Success = next-token likelihood	Success = closed grammar + certified commit + joule receipt
Model is the substrate	Model is a demoted residual leaf

Clean-room policy: Mixture of Limits is law + runtime, not a wrapper on MoE, and not a path-dependent fork of sibling OpenIE trees for the proven path.

3. Proof spine: $E(x) \geq \theta(D) \cdot \mu(S, V)$

The energy / impedance law used throughout Mixture of Limits is:

$$E(x) \geq \theta(D) \cdot \mu(S, V)$$

where $E(x)$ is path energy for closing act x , $\theta(D)$ scales with decision / difficulty structure D , and $\mu(S, V)$ is an impedance factor for stack coordinate S and view V . Soft-ref receipts stamp catalog μ with `mu_source=catalog` and a `landauer_floor_ratio`. The product is an **analytical / catalog estimate**, not a package joule reading.

Landauer (estimate only). For irreversible erasure of n bits at temperature T ,

$$E_{\text{Landauer}}(n, T) = nk_B T \ln 2$$

Receipts may annotate `landauer_floor_J` from this formula. That annotation is **estimate ≠ measured**. Soft-ref closes keep `measured_j=None`. Optional OS meter features (RAPL / IOReport / powermetrics), when present and successful, may populate `measured_j` only under a labeled `MeasureSource`; failure or VM stays unavailable. Soft-ref prove criteria do not require those meters.

Historical rhyme. Kepler did not need every future observation once the three laws closed the grammar of planetary motion for the epoch. Newton did not need a larger ephemeris table to predict a new orbit once $F=ma$ and inverse-square gravitation covered the coordinate. Shannon did not need infinite samples to bound channel capacity. Landauer did not need a particular chip to state a thermodynamic lower bound. Mixture of Limits spine is the same move for CI: bind a floor, close cheapest-sufficient, refuse when the floor says stop.

Constructive existence (software). Soft-ref cargo `run -p mol-cli -- prove` prints VERIFIED per criterion and exits 0 (~29 soft-ref criteria spanning deterministic close, formula/lookup without model, Vol refuse, settle commit+refuse, certificate refuse, capability default-deny, receipt labels, replay-class coercion deny, Periodic Stack subset navigation, μ catalog, transcript replay, Z2 cite / Z1 compose, agent mailbox, bitemporal memory, fabric routing, desktop headless shell, OS meter labels, WASM capsule, Agent Lane, multi-fabric receipts, ecosystem e2e certify; see PLAN.md in the Mixture of Limits workspace). That is constructive existence **in software**—a runtime fact, not a board energy measurement. Prove↔claim mapping is Leapfrog-owned (§6).

4. Cascade and close: Lookup → Formula → Solver → Model LAST

MathGround / OpenIE cascade order (type-enforced replay classes; synthesis zones and compute primitives on synthesis.openie.dev and compute.openie.dev):

```
Lookup → Formula (closed-form) → Solver / settle → Model LAST
```

Mixture of Limits tier	Role	Replay class (typical)
Lookup	LUT / allow-table / registry hit	Deterministic
Formula	Closed identity / unit convert / law instance	Deterministic
Solver	Sparse solve / ternary settle	Deterministic or compose under Solver
Model	Stochastic residual only	ModelGenerated

Invariant: ModelGenerated →/Deterministic. Typed answers refuse coercion.

Automation / close loop:

```
sense → classify grammar → cheapest sufficient → certify → commit|refuse → receipt
```

Mixture of Limits **owns close**. Refuse over escalate when unsafe or Vol-negative. Refuse is a **lawful success** and still emits a receipt.

co/0 bind sketch (namespaces)

Soft-ref receipt labels bind act namespaces in the spirit of OpenIE co/0 receipts (sketch, not a live leapfrog path-dep):

Namespace	Role
claim.*	Seeded / retrieved knowledge claims (Z2); unknown → refuse, no invent
act.*	Capability-gated acts; Mutate/External default-deny
close.*	Route + certify gate; commit or refuse
receipt.*	estimated_j labeled; measured_j only if Metered probe returned; board_synth_claimed=false

Rules stamped on every soft-ref close:

1. **Certify-before-commit**; no irreversible side effect without certificate pass.
2. **ModelGenerated** →/Deterministic; sealed replay markers.
3. **measured_j only if Metered**; otherwise None / unavailable.

4. **Refuse = lawful success + receipt.** Vol / settle_refuse / certificate / capability denies still close the ledger.

Multi-fabric soft-ref: after tier selection, route cheapest sufficient DeviceKind (Cpu always present; Gpu*/Wasm/ThermoSettle/... optional). Detection ≠ joules.

5. Econ bridge: Satiation

Digital inference prices can fall toward free-at-margin without implying free energy, free actuation, or unbounded value after a chore is complete. The companion study [Satiation and Scarcity after Free AI](#) defines satiation as a stop on a written completeness predicate and cites published inference-price series. Mixture of Limits supplies the *machine* stop that matches that economic intuition: Vol and grammar floors refuse excess generation once benefit is covered; commit/refuse receipts (NI) record the irreversible boundary.

Satiation owns Economic Reality of Satiation and its price tables; NI owns the commit record; Mixture of Limits owns navigation floors that make “done” and “unsafe” executable without opening the model leaf.

6. Prove ↔ claim map

The Mixture of Limits constructive prove is not a leaderboard stunt. It is an **information-theoretic** close law rooted in **physics**: the same historical arc that took astronomy from excess epicycles to Kepler/Newton predictive floors, and that takes computation from Shannon → Landauer → complexity / Vol toward named energy floors. Global academia already holds those proof points; the bottleneck is **applied math × materials (hardware) embodiment**, not awareness of the slogans. This section only maps what mol prove actually stamps VERIFIED onto paper claim.* ids: embodiment of known floors, not invention of new physics.

A claim is **proven** only when the in-tree harness prints VERIFIED for the named criterion and exits 0. A claim is **soft-ref only** when PLAN §6 marks it OUT OF PROOF SCOPE, or when the criterion itself stamps soft-ref / fixture / feature-gated behavior rather than live silicon. Proven-path rows keep measured_j=None (or Metered only when a probe returns) and board_synth_claimed=false.

Measurement labels used below (OpenIE measurement grammar):

Label	Meaning
Unmetered	Software-ref / soft-ref path; measured_j=None; estimate may be labeled
Estimated	Catalog / Landauer / fuel / μ surrogate with explicit EstimateKind or mu_source=catalog
Metered	Optional live probe (energy-meter) returned numbers (not required for prove)

Default close path for all proven claims: **Unmetered** (+ **Estimated** where μ / Landauer / fuel appear).
board_synth_claimed=false everywhere in the proven path.

6.1 Harness meta

Prove #	Criterion (PLAN)	Paper claim id	Status	Measurement
P0	cargo test -- workspace green	claim.mol.ci_workspace	Proven (CI / operator; not printed by mol prove)	-
P1	cargo run -p mol-cli -- prove exits 0	claim.mol.prove_harness	Proven (prints VERIFIED per row below)	Unmetered

6.2 Proven claims (P2-P24)

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
P2	Deterministic close (double-run)	claim.mol.deterministic_close	Same asks yield the same commit/refuse + limit id.	-	Un
P3	Formula/lookup close without model	claim.mol.formula_lookup_no_model	Landauer / unit-convert closes COMMIT at Lookup or Formula; model never answered.	-	Un Est (La floc rec
P4	Vol refuse when !allow_model	claim.mol.voi_refuse	Free-form ask → REFUSE voi.	-	Un
P5	Settle commit + settle refuse	claim.mol.settle_ternary	Ternary settle → COMMIT; will not settle → REFUSE settle_refuse.	Software-ref settle stub; no klerem / FPGA	Un
P6	Certificate refuse on diverge	claim.mol.efa_certificate_refuse	Formula + diverge → REFUSE efa_certificate.	In-tree EFA-style cert; Ferric / live BMI loop soft-ref	Un
P7	Capability default-deny mutate	claim.mol.capability_deny_mutate	AutomateGate::default().gate(Mutate) → Refuse.	-	Un

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
P8	Receipt labels	claim.mol.receipt_honesty	Software-ref: measured_j=None, board_synth_claimed=false, estimate labeled (EstimateKind).	Live RAPL/ NVML default close OUT OF SCOPE	Unr Est
P9	ModelGenerated →/ Deterministic	claim.mol.replay_no_strengthen	TypedAnswer::weaken_to returns ReplayCoercion.	-	-
P10	Periodic Stack live catalog navigation	claim.mol.stack_live_catalog	Family + present + scale → COMMIT at Lookup; 258 live Lookup/Formula/Solver/ Navigate gears; note cites 258/33.	μ calib + silicon HW OUT OF SCOPE; soft-ref Gap sims in proof	Unr
P11	primitive_gap via registry probe	claim.mol.primitive_gap	Gap marker physical_settle + absent name → REFUSE primitive_gap (not string-only).	-	Unr
P12	μ / impedance catalog	claim.mol.mu_catalog	Receipts stamp mu_source=catalog, mu, landauer_floor_ratio; $E \approx \theta \cdot \mu$ (catalog estimate, not RAPL/measured_j).	μ calib corpus OUT OF SCOPE	Est
P13	Receipt transcript replay	claim.mol.receipt_replay	JSONL/in-memory replay reproduces commit/refuse + limit id; model never answered.	-	Unr
P14	Z2 retrieve+cite	claim.mol.z2_retrieve_cite	Seeded factual hit → COMMIT RetrievedCited + citation ids; unknown → REFUSE claim_unknown (no invent).	Seed corpus in-tree only	Unr
P15	Z1 compose/ synthesis	claim.mol.z1_compose	≥2 cited claims → COMMIT Composed + synthesis.composed_from; missing → REFUSE compose_missing; never laundered as Deterministic/RetrievedCited alone.	-	Unr
P16	Agent mailbox loop	claim.mol.agent_mailbox	Goal/Message/Act → close → commit/ refuse + receipt in transcript; demo hits formula/cite/compose COMMIT and voi/	-	Unr

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
			settle_refuse REFUSE; model cold; measurement flags hold.		
P17	Bitemporal state + memory	claim.mol.bitemporal_memory	Writes only on Mixture of Limits close COMMIT; Mutate deny / free-form remember → REFUSE; recall → RetrievedCited + memory: cite; unknown → memory_unknown.	-	Un
P18	Fabric routing + soft-ref inventory	claim.mol.fabric_soft_ref	Offline inventory valid (source=software_ref, Cpu always); Lookup/Formula→Cpu; Settle→ThermoSettleCpu; Model→Gpu* or REFUSE fabric_unavailable; detect ≠ measured joules.	Live wgpu presence optional / not required for prove	Un
P19	Desktop shell headless	claim.mol.desktop_headless	ShellSession ask/close → receipt view (zone/fabric/limit/estimated_j/ measured_j=None); joule ledger; Vol refuse; GUI not required.	egui GUI optional	Un Est
P20	OS meter labels	claim.mol.os_meter_honesty	Feature off / VM / no permission → measured_j=None, empty components, measure_source=unavailable; fixtures stamp labeled sources; default close stays unmetered.	Live probe optional (energy-meter)	Un (de Me wh retu
P21	WASM capsule certify	claim.mol.wasm_capsule	add.wasm under stub runtime; fuel → estimated_j only; FS/net grant / fuel exceed / host-share → REFUSE; close stamps capsule.	Optional wasmtime feature	Un Est (fue
P22	Agent Lane session path	claim.mol.agent_lane	Lane partitions; host_invoke without provenance refuses; keyword confirm insufficient; need GrantReceipt; allow stamps AgentLaneReceipt.	-	Un
P23	Multi-fabric compute receipt	claim.mol.fabric_compute_receipt	Soft-ref inventory; ComputeFabric / ScheduleDecision / ComputeStepReceipt with fabric_id; cascade stamps fabric:route; CPU Formula COMMIT; Model residual may fabric_unavailable; measured_j=None on soft-ref path.	Ferric not path-dep'd	Un
P23b	Live fabric soft	claim.mol.fabric_live_soft	Feature off stays software_ref; mock/live Metal can stamp gpu_metal; Formula stays Cpu; missing Gpu* fail-closed; prove	Soft / optional detect	Un

Prove #	Criterion (PLAN)	Paper claim id	Claim (one sentence)	Soft-ref / scope note	Me
			does not require a GPU; measured_j=None unless Metered.		
P23c	wgpu / Metal tiny kernel	claim.mol.kernel_vector_add	Clean-room WGS� vector-add; soft stub offline (same checksum); Gpu* stamps kernel:vector_add + execution_proof; optional overlapping meter may stamp package measured_j with energy_honesty=measured (field name); no rail-sum package joules.	Live Metal optional; fixture OK for prove	Un (de Me if m retu
P24	Ecosystem e2e certify (one receipt)	claim.mol.ecosystem_e2e	Lane → fabric (Cpu soft-ref) → WASM → energy labels (estimated fuel; optional meter_sample) → host invoke only with GrantReceipt → commit/refuse on one receipt; refuse-without-grant → grant_receipt_required.	-	Un Est

6.3 Soft-ref only

These are PLAN §6 residuals. Citing them is allowed as roadmap / reference semantics. Treating them as measured results is not.

Soft-ref item	Why soft-ref	Related prove row (if any)
Live silicon RAPL / NVML as default close	Optional energy-meter; soft-ref prove stays measured_j=None	P8, P20
Live wgpu adapter presence	Feature fabric-detect; inventory soft-ref remains prove default	P18, P23b, P23c
Ferric / on-device EFA (BMI hardware loop)	Reference semantics only; not path-dep'd	P6, P18, P23*
klere-vm WASM / FPGA meter (real pJ)	Software-ref StubKlereSettle	P5
Live openie-path / leapfrog ask bridge	Adapter port	-
Live WCA MCP / wca-lut-edge in-proc certify	Adapter port	-
μ calib corpus + silicon HW (physical_settle / reversible / ising /	Live catalog (258 Present / 258 live gears) + Gap + tier μ catalog + Stage C/Ferric soft-ref inventories + HW Gaps	P10, P12

Soft-ref item	Why soft-ref	Related prove row (if any)
adiabatic / ferric / quantum / analog / photonic Gap cells)	soft-ref sims x8 (stage_c_measured=false; Gap cells retained) are in proof; silicon meters OUT	
Trained weights / candle / tract Model leaf	Model stays demoted stub	P3, P4, P16

6.4 Seeded knowledge claims (Z2 corpus; not the prove map)

In-tree ClaimStore seeds used by P14/P15 (retrieve/compose), distinct from the paper claim ids above:

- claim:landauer.principle
- claim:mol.law
- claim:mol.proof_law
- claim:mol.cascade
- claim:stack.scale
- claim:honesty.no_fake_rapl
- claim:replay.retrieved_cited
- claim:openie.z2

6.5 Measurement scope of the proven map

- Soft-ref prove path: measured_j=None (Unmetered); Metered only when a live probe returns.
- board_synth_claimed=false on the proven path.
- Soft-ref v0.1 prove is software constructive existence, not board / FPGA watt leadership.
- Soft-ref roadmap items (§6.3) stay roadmap; they are not escalated to proven claims.

7. Math sketches: Vol stop, grammar coverage, settle-refuse

7.1 Vol stop

Let B be a benefit functional for a decision, x the current information state, and c the cost (bits, latency, or estimated joules) of acquiring observation y. A simple stop rule:

$$\Delta(x, y) = \mathbb{E}[B \mid x, y] - \mathbb{E}[B \mid x] - \lambda c(y)$$

Refuse escalation when $\Delta(x, y) \leq 0$ (or $\leq \tau$ for a configured threshold). Soft-ref Mixture of Limits encodes this as floor voi with allow_model=false by default: free-form asks that would only open the model leaf refuse rather than spend. This is the Shannon/decision-theoretic cousin of Kepler stopping new naked-eye points once the law predicts within tolerance. Domain-calibrated Bayesian Vol tables remain a deployment parameter, not a soft-ref prove deliverable. Classical cousins:

anytime algorithms allocate deliberation under a utility-of-computation schedule (Zilberstein, 1996); the **information bottleneck** prices compressed representations that keep task-relevant bits (Tishby, Pereira, and Bialek, 1999; Tishby and Zaslavsky, 2015, arXiv:1503.02406). Soft-ref Mixture of Limits encodes the stop as named floors rather than importing those libraries as prove deliverables.

7.2 Grammar coverage

Let G be the covered grammar (Lookup registry $\cup$ Formula identities $\cup$ Solver settle programs). A query q is **covered** if a deterministic derivation $q \vdash_G a$ exists with replay class not `ModelGenerated`. Cascade order tries Lookup, then Formula, then Solver. If covered, **do not open the model**. If uncovered and policy forbids model escalation, refuse (`grammar / voi / primitive_gap` as applicable). Historical analog: once Newton's grammar covers orbital mechanics for the force law at hand, one does not re-enumerate Brahe's table to answer a new initial-condition query.

7.3 Settle-refuse

Under Solver, a ternary / energy-landscape settle either reaches a certified fixed point (`commit`) or signals `will-not-settle` (`refuse settle_refuse`). Soft-ref implements this as real in-tree settle logic plus certificate diverge tags (`efa_certificate`). Thermodynamic-settle inspiration is conceptual: soft-ref v0.1 has no annealer drivers and no adiabatic wall joules. Landauer appears as a labeled estimate on receipts.

7.4 Bound sketch tying Shannon and Landauer

For n irreversible erasures on a path that closes a decision,

$$E_{\text{path}} \geq E_{\text{Landauer}}(n, T) = nk_B T \ln 2$$

and the Mixture of Limits catalog estimate further requires $E_{\text{est}}(x) \geq \theta(D) \cdot \mu(S, V)$. `measured_j` requires a Metered probe. Excess tokens that do not change B fail the Vol test even when joule budgets remain; the information floor binds first.

8. Related work: proof points, lineage, and Tier A

8.0 Academic proof points: top minds already named the floors

Mixture of Limits is an executable navigation law over floors that information theorists, physicists, and decision theorists already stated as settled theory, and that formula-first science AI keeps rediscovering in modern form. Below is a **historical** $\rightarrow$ **recent citation chain** with DOIs verified for this draft (or marked TBD). These are global academic proof points that compression, floors, and formulas beat excess enumeration.

Information, complexity, and physical cost

Proof point	Who / when	What they already said	Cite (verified)
Bits under uncertainty	Shannon (1948)	Quantitative communication limits; entropy as missing information	10.1002/j.1538-7305.1948.tb01338.x (Part I); Part II 10.1002/j.1538-7305.1948.tb00917.x
Irreversible erasure costs heat	Landauer (1961)	Logical irreversibility $\Rightarrow$ minimal heat $\sim kT$ per irreversible function	10.1147/rd.53.0183
Reversible computing	Bennett (1973)	Computation can be logically reversible; Landauer cost is about erasure, not “computing” per se	10.1147/rd.176.0525
Algorithmic complexity	Kolmogorov (1965/1968)	Quantity of information as shortest program length; compression as definition	Russian original <i>Probl. Peredachi Inf.</i> 1(1):3–11 (1965), no DOI; English transl. 10.1080/00207166808803030 ; IEEE note 10.1109/TIT.1968.1054210
Algorithmic probability / induction	Solomonoff (1964)	Prefer short programs; universal prior over computable regularities	Part I 10.1016/S0019-9958(64)90223-2
Algorithmic incompleteness	Chaitin (1977)	Algorithmic information bounds what finitely axiomatized theories can prove	10.1147/rd.214.0350
Probability as extended logic	Jaynes (2003 book; earlier papers)	Maximum-entropy / logic-of-science discipline against ad-hoc excess modeling	10.1017/CBO9780511790423
Textbook IT + decision links	Cover & Thomas	Canonical Elements of Information Theory (2nd ed.)	Wiley book DOI 10.1002/047174882X
Value of Information	Howard (1966)	Shannon bits $\neq$ decision value; Vol joins probability with economic consequence; stop when information is worthless for the decision	10.1109/TSSC.1966.300074
Physics of computation lectures	Feynman (pub. Hey/Allen eds.)	Physical limits of computing, reversible/thermo themes in lecture form	Anniversary ed. 10.1201/9781003358817 ; 1996 Addison-Wesley ISBN 0-201-48991-0

Historical mathematical physics (compression beats enumeration). See §8.1:

Brahe→Kepler→Newton is the pre-IT proof that tables yield to laws. Imprint DOIs for critical editions of *Principia* remain **TBD** if a journal version requires them.

Recent formula / mechanism / certify systems (Tier A inventory), from docs/adjacent-field-hunt.md and docs/appendix-mwm-ai-newton.md (2026-09-30):

Proof point	Who	Mixture of Limits reading	Cite (verified)
AI Feynman	Udrescu & Tegmark	NN decomposes; closed form is the commit	10.1126/sciadv.aay2631
SINDy	Brunton / Proctor / Kutz	Sparse grammar; reject dense residual fits	10.1073/pnas.1517384113
PySR	Cranmer	Production symbolic regression	arXiv: 2305.01582 (journal DOI TBD if distinct)
AI-Newton	Fang et al.	Concept library + general laws across experiments	arXiv: 2504.01538
AlphaGeometry / AlphaProof	DeepMind	Propose → symbolic/Lean certify (hard floor)	Nature 10.1038/s41586-025-09833-y
LeanDojo / ReProver	Yang et al.	Retrieval-augmented Lean proving; tool interaction with proof state	arXiv: 2306.15626
Mechanistic World Models	Posner / Lei / Schölkopf	Mechanisms > predictive MoE modules	arXiv: 2607.12474
AutoSINDy hybrid	(hunt 2026 note)	PySR → library → SINDy	arXiv: 2605.09696

Raiffa-style decision analysis texts are the pedagogical cousins of Howard's Vol; a specific Raiffa imprint DOI is **TBD**.

8.0.1 The gap is embodiment: not awareness

Claim (addendum). Applied mathematics × materials science (physical hardware) is the bottleneck, **not** awareness of what should be done.

Evidence for the awareness side is the table above: Shannon priced bits; Howard priced Vol for decisions; Landauer/Bennett priced irreversible erasure; Kolmogorov/Solomonoff/Chaitin priced shortest programs and incompleteness; Feynman taught physical limits of computation; SINDy/PySR/AI Feynman/AI-Newton/AlphaGeometry/MWMs keep showing that formulas, sparsity, verification, and mechanisms beat unbounded enumeration and unguided MoE-style prediction.

What remains hard, and what Mixture of Limits companions (NI commit law; coin-cell Formula+LUT; multi-fabric routing; Metered vs Estimated labels) point at, is **embodying** those floors in real stacks: materials and devices that make Lookup/Formula/Solver cheap, certificates enforceable before

irreversible acts, and joule receipts that keep estimates separate from measured_j. Soft-ref mol prove shows the *law+runtime* exists in software. Materials and device embodiment remain open. Soft-ref path: board_synth_claimed=false; package energy only when Metered.

8.1 Historical lineage: compression beats enumeration

Tycho Brahe → Kepler. Brahe's program was high-fidelity enumeration. Kepler's three laws compressed those tables into predictive relations. Enumeration was necessary input; the law was the compressible object that stopped needing every future observation.

Kepler → Newton. Newton folded planetary regularities into invariants and closed forms ($F=ma$; universal gravitation). Predictive power migrated from case tables to formulas that travel. AI-Newton (Fang et al., arXiv:2504.01538) is a contemporary soft echo: concept libraries and general laws across many noisy mechanics experiments. NN as recommender residual, not substrate.

Invariants and closed forms. Symmetry → conservation (Noether tradition), Hamiltonian/Lagrangian reduction, and integrals of motion are the mathematical-physics statement of "formula wins when grammar is covered." The Mixture of Limits Formula gear is that tradition's CI leaf.

Information theory and physical cost. Shannon, Howard (Vol), Landauer, Bennett, Kolmogorov/Solomonoff/Chaitin, and Cover & Thomas form the IT/decision/thermodynamic chain in §8.0. Mixture of Limits binds them as named floors and labeled estimates. Soft-ref Landauer stamps are lower-bound estimates, not claims of near-bound silicon operation.

Notational intelligence. Lee's notational-intelligence claim is the companion frame for executable predicates; see [Notational Intelligence as Commit Law](#).

8.2 Tier A detail: formula / mechanism / executable physics

Inventory dated 2026-09-30 in the Mixture of Limits workspace hunt docs. Capsule readings:

1. **AI Feynman (+ 2.0):** Physics-inspired symbolic regression; NN for symmetries/separability, then recursive formula recovery. Canonical *Formula* gear.
2. **SINDy:** Sparse identification of nonlinear dynamics; explicit grammar + sparsity floor.
3. **PySR:** Evolutionary symbolic regression; distill NNs → closed forms.
4. **AI-Newton:** Concept-driven law discovery across noisy mechanics experiments.
5. **AlphaGeometry / AlphaProof:** LM proposes; verifier is the hard floor (propose → certify → commitlrefuse template).
6. **Mechanistic World Models:** Sister philosophy: reusable mechanisms; MoE as wrong module semantics for explanation.

Additional hunt rows (AtomAgents/SciAgents, KeplerAgent/NewtonBench, AI-Descartes, DreamCoder/COMET/NEO, Robot Scientist) sit as hybrid or mechanism-library cousins. Where a DOI was not verified beyond arXiv IDs, cite arXiv and mark journal DOI **TBD**.

Mixture of Limits reading of Tier A. Formula/mechanism discovery (A) plus physics-executable leaves (B in the hunt) beat token-escalation workflows (C). MoE-style mixtures are the wrong “Mixture of.”

8.3 What Mixture of Limits adds

Tier A systems recover or verify formulas. Classical IT already priced bits, Vol, and erasure. Mixture of Limits names the **navigation law** that decides *when* Lookup/Formula/Solver suffice, *when* to refuse, and *how* receipts label estimated vs measured joules, then implements a clean-room soft-ref close loop. Soft-ref constructive existence is software constructive existence under those floors—not a materials breakthrough or a new physics corpus.

8.4 Latent space: depth buys compression, not freedom from floors

A **latent space** is a compressed coordinate system for states that matter to a task: continuous vectors (dense embeddings), discrete codes (VQ / codebook indices), or hybrids. Depth in a latent buys what Shannon and Kolmogorov already price: a shorter description of structure that would otherwise be enumerated as raw observations, pixels, tokens, or traces. Autoencoders, VAEs, diffusion latents, and continuous reasoning traces are modern instances of that move. The representation can make Lookup hits cheaper, Formula edits local, and Solver landscapes smoother.

Representation is not law. A deeper latent does not erase the floors Mixture of Limits names:

1. **Value of Information (Vol).** Extra latent dimensions, longer continuous traces, or denser codebooks are more bits under a budget. When marginal latent bits do not change the decision benefit B, Vol says stop. Excess latent enumeration is the same failure mode as excess token enumeration, only in a different alphabet.
2. **Grammar coverage.** If Lookup, Formula, or Solver already covers the coordinate, opening a latent generator is waste. Covered grammar closes before Model. A latent that rediscovers a closed form is still Model residual when a Formula leaf was available.
3. **Energy / Landauer.** Encoding, decoding, denoising, and gradient edits on latents erase and rewrite information. Landauer prices irreversible erasure as a thermodynamic lower bound and appears on receipts as a **labeled estimate**. Catalog or OpCounter estimates for latent paths remain **Estimated / Unmetered** on the soft-ref path (`measured_j=None; board_synth_claimed=false`).
4. **Certify-before-commit.** A low-energy latent state is a proposal until a certificate, settle, or typed refuse closes the act. Soft-ref close still binds `propose → certify → commit|refuse → receipt`. Latent score $\neq$ irreversible commit.

Teach-first takeaway. Latent depth is a compression technology and a plateau option when Model is open. Mixture of Limits navigates whether that compression is cheapest-sufficient for the task, or whether refuse is the lawful success. See §8.7 (SOTA at the plateau) and §9: the question is whether a generator should run at all, then how little energy to spend on the plateau.

8.5 Logical Intelligence: energy-based reasoning with LLMs and latents

Logical Intelligence (logicalintelligence.com) is a company building energy-based reasoning systems for constraint-heavy and mission-critical settings. Public materials (company blog and product pages, January 2026) teach a hybrid stack rather than a single chatbot:

- **Kona** is their core **energy-based model (EBM)** for reasoning (sometimes called an energy-based reasoning model, EBRM). An EBM assigns a scalar **energy** to a candidate state: low energy means more consistent with constraints and objectives; high energy means something is broken. Per their technical blog (Bodnia and Hanin, 21 Jan 2026), Kona is non-autoregressive at the *trace* level, globally scored over partial and complete traces, and reasons in a **continuous latent space** with dense vector tokens so local gradient-style edits can reduce constraint violations without regenerating an entire discrete prefix.
- **Aleph** is their orchestration / agentic layer that coordinates Kona, LLMs, and other tools. Public positioning: LLMs handle natural-language interface and candidate generation; the EBM layer evaluates and repairs under constraints. Aleph is also described as delivering verified / formal-reasoning workflows today (benchmark claims such as PutnamBench appear on their site; treat score numbers as **author-reported** unless independently reproduced here).
- Product framing on the Kona page: Kona is not marketed as a chatbot. Language models express and explore; Kona is positioned to evaluate what is valid or permissible before irreversible action in high-stakes domains (company examples include energy, manufacturing, semiconductor verification, robotics). Kona 1.0 was announced for partner pilots (company and press materials, Jan 2026). Leadership publicly listed includes founder and CEO Eve Bodnia and Yann LeCun as founding chair of a technical research board (company site / press).

Classical EBM backdrop (settled ML architecture). Energy-based models treat inference as finding low-energy configurations under a learned energy function. That framing is older than any one product; LeCun and others have long argued that reasoning can be cast as optimization over an energy landscape. Logical Intelligence's public thesis is that discrete, locally scored, autoregressive LLM traces scale poorly for long-horizon constraint satisfaction, and that continuous, globally scored EBM traces address that gap when paired with LLMs for interface.

Mixture of Limits reading (cascade map, not a product endorsement).

Logical Intelligence public piece	Cascade rhyme	Floor that still binds
Continuous latent trace + energy score	Soft rhyme with Solver / settle : optimize / repair under constraints	settle_refuse, Vol, energy estimate; low energy $\neq$ board joules
LLM for language / candidates	Model residual for expression and proposal	Model LAST; ModelGenerated $\rightarrow$ / Deterministic
Aleph orchestration among tools	Cheapest-sufficient routing among gears	Grammar coverage first; refuse when escalation is Vol-negative or unsafe
	Certify-before-commit spirit	

Logical Intelligence public piece	Cascade rhyme	Floor that still binds
Constraint / proof-oriented close		Certificate / typed refuse before irreversible act

Verified here: the public architecture story and the cascade map—energy-based settle can rhyme with Solver; LLM residual stays Model LAST; latents do not erase Vol or Landauer floors. Author-reported latency and Sudoku figures stay field results (not soft-ref measured_j). Soft-ref path: board_synth_claimed=false. Unverified internal training details, unpublished energy-function forms, and AGI-completion claims remain **TBD** / out of scope.

8.5.1 Embodiment classes: the convergence is hybrid interplay

§8.5 already separates the objects inside one public stack. The energy-based settle returns a low-energy configuration. The scalar over a trace, as §8.5 cites, is constraint energy. It is not a measured joule. The language model beside it returns candidates. Neither object is the commit. Close remains propose, then certify, then commit or refuse, then a receipt (§4). The same separation is the limit of each embodiment class. It is not a feature of one name.

A **decision model** stops at a typed probability. The return is a yes or no, a choice, or a score, each with a probability. Whatever route, escalation, or deferral follows is outside that return. The model does not commit. A probability is not an act.

A **bounded ask** stops at a typed value, or at not-sure. The ask writes no text and takes no action. Not-sure is a refuse object, not a generated answer.

A **statechart** stops at a legal transition. An event changes state only when that state allows it. An illegal event does not transition. The object returned is the next legal state, or no transition. A library transition is not an external commit.

A **visual statechart editor** stops at the chart a person can audit. The editor's object is the chart. The chart is the door: what the chart does not allow does not pass. The editor does not commit the act behind the door.

An **energy-based settle**, already placed in §8.5, stops at a low-energy configuration. Constraint energy is not measured_j. A low-energy configuration is not an irreversible act.

None of these classes is the whole intelligence. The convergence is hybrid interplay. The chart is the door. The decision model may fill a proposal. The bounded ask may return a typed value or not-sure. The energy settle may score a configuration. Something outside the model still commits or refuses. Mixture of Limits is the navigation law of that interplay: Lookup → Formula → Solver → Model last. Refuse is success. Energy to run is the only true metric. All other factors collapse to zero. Constraint energy remains a score on a configuration. It is not that metric.

A shared request shape is not shared authority. Two typed answers still do not commit. These stops are not plateau knobs inside one generator (§8.7). Stacking classes does not produce a commit. Soft-ref path: `board_synth_claimed=false`. Nothing in the class map is a measurement of this study.

Class map

The class map places each public instance under the object that instance returns. Rows are ordered by class. A project that returns two objects appears twice, once for each object. The map is not a ranking.

Class	Object returned	Instance
Decision model	Typed probability	Jev
Decision model	Typed probability	Clef
Decision model	Typed probability	d1
Decision model	Typed probability	Solar Decide
Decision model	Typed probability	pplx-decider
Decision model	Typed probability	Kev
Decision model	Typed probability	Laya
Decision model	Typed probability	Bespoke Nimble
Decision model	Typed probability	OpenThai-SystemOne
Decision model	Typed probability	Decider
Decision model	Typed probability	OpenJev
Decision model	Typed probability	Mica
Decision model	Typed probability	Winnow
Decision model	Typed probability	CLM
Decision model	Typed probability	Julia
Decision model	Typed probability	NanoJev
Decision model	Typed probability	open-jev DeBERTa
Decision model	Typed probability	fastText
Decision model	Typed probability	Cloud Natural Language
Bounded ask	Typed value, or not-sure	ThinkThen
Bounded ask	Typed value, or not-sure	GLiNER2.5-Decide
Statechart	Legal transition	XState

Class	Object returned	Instance
Statechart	Legal transition	Qt State Machine
Statechart	Legal transition	Qt SCXML
Statechart	Legal transition	Boost.Statechart
Statechart	Legal transition	Boost.MSM
Statechart	Legal transition	Boost.SML
Statechart	Legal transition	Akka FSM
Statechart	Legal transition	Spring Statemachine
Statechart	Legal transition	Apache Commons SCXML
Statechart	Legal transition	transitions
Statechart	Legal transition	Sismic
Statechart	Legal transition	Stateless
Statechart	Legal transition	QP/C
Statechart	Legal transition	AASM
Statechart	Legal transition	SCION
Statechart	Legal transition	javascript-state-machine
Statechart	Legal transition	state_machines
Statechart	Legal transition	HFSM2
Statechart	Legal transition	uSCXML
Visual statechart editor	Auditable chart	Stately
Visual statechart editor	Auditable chart	itemis CREATE
Visual statechart editor	Auditable chart	QM
Visual statechart editor	Auditable chart	Engineering Rhapsody
Visual statechart editor	Auditable chart	Enterprise Architect
Visual statechart editor	Auditable chart	Papyrus
Visual statechart editor	Auditable chart	StarUML
Visual statechart editor	Auditable chart	Visual Paradigm
Visual statechart editor	Auditable chart	IAR Visual State
Visual statechart editor	Auditable chart	sketch.systems
Visual statechart editor	Auditable chart	Qt Creator
Energy-based settle	Low-energy configuration	Kona

Class	Object returned	Instance
Energy-based settle	Low-energy configuration	Simulated Bifurcation Machine
Energy-based settle	Low-energy configuration	Fixstars Amplify AE
Energy-based settle	Low-energy configuration	CMOS annealing

8.6 World Labs: spatial intelligence and world models

World Labs (worldlabs.ai) is a frontier research and product company focused on **spatial intelligence**: models that perceive, generate, reason about, and interact with virtual and physical worlds across space and time. Co-founders publicly include Fei-Fei Li, Justin Johnson, Ben Mildenhall, and Christoph Lassner (company About page).

Teach the products and architecture first (public sources).

- **Marble** is their first product: generative 3D world models that create spatially coherent, persistent 3D worlds from images, video, text, and 3D layouts (company About).
- **Atlas** (company blog, 1 Sep 2026) is described as an omni **world model** pretrained to operate on text, images, video, and 3D. Architecture: a **multimodal autoregressive diffusion transformer**. Inputs are grounded in 3D to form a shared **spatial context**; the model generates what comes next while aiming for 3D consistency with what it has seen, and imagining what lies beyond. Public capability claims include camera-controlled video generation, sparse-view spatial reconstruction (including explicit 3D such as point clouds / Gaussian splats), space-time simulation for Real-to-Sim robotics workflows, and image / 360 generation. Atlas is positioned to power future Marble versions and is in early access with select partners. Author-reported benchmarks on camera-conditioned generation and 3D reconstruction appear on the Atlas post; cite the architecture and task list here, not those numbers as OpenIE measurements.

Fei-Fei’s public spatial-intelligence framing (including the “from words to worlds” thesis) states that world models must handle semantic, physical, geometric, and dynamic complexity beyond today’s LLM text corridor. That is a generator-class ambition: better world generators and simulators.

Mixture of Limits reading: even projected-superior world models hit floors.

1. **Grammar coverage.** Faithful reconstruction from enough views is not the same as a closed predictive law for a task. When a Formula or Lookup already answers the coordinate, generating a world is excess. Imagination that fills unseen regions is generative residual, not Deterministic commit.
2. **Certify-before-commit.** A spatially consistent video or splat is a proposal about geometry and appearance. Irreversible acts (robot motion, financial or safety-critical side effects) still require certify → commit/refuse → receipt. Soft-ref ReplayClass typing still applies: generative world output does not launder into Deterministic without a certificate path.

3. **Vol and energy.** Longer videos, denser 3D contexts, more denoising steps, and Real-to-Sim rollouts spend bits and joules. Vol stops when extra world detail does not change the decision benefit. Landauer bounds irreversible erasure as a labeled estimate; Atlas / Marble training or inference joules are field / author-reported, not soft-ref measured_j. Soft-ref path:
board_synth_claimed=false.
4. **Materials embodiment.** High-fidelity world models intensify the applied-math × materials bottleneck: sensors, fabrics, and meters that make cheapest-sufficient close real at the edge, without laundering estimates as board watts.

World Labs systems optimize **which world generator / simulator runs** and how well it tracks space. Mixture of Limits asks whether that generator should run for the stated benefit, and which named floor closes first. Cross-link: §9 Mixture of Limits is not MoE table (outside-generation floors vs inside-generator capacity).

8.7 SOTA at the plateau: Vol zero past the floor

Plateau thesis. Once the floor is known and Value of Information is zero past it, additional tokens, parameters, latent depth, or thinking steps do not purchase verifiable progress on the task coordinate. The only remaining work is **optimize energy and compute on that plateau**: cheapest-sufficient Lookup → Formula → Solver/settle → Model LAST, with refuse as lawful success when escalation is Vol-negative or unsafe.

Latent compression, EBM settle hybrids (Logical Intelligence), spatial world models (World Labs), test-time compute, super learning, SSM hybrids, mixture of experts, speculative decode, RAG, transmission / gearing / cascade / memory-context methods, and the cross-field plateau moves of §8.14 can each be projected as superior *inside* a generative or hybrid corridor for their stated jobs. Superiority on the plateau does not cancel Shannon pricing of bits, Howard pricing of Vol, Landauer pricing of irreversible erasure, or certify-before-commit. Those methods are **implementation options for plateau spend**—how to spend less energy or move fewer bits once Model is the residual leaf—not history lessons and not a narrative race among brands. The industry race is a race to the plateau floors.

Mixture of Limits is the **navigation law** that names the floors, finds the plateau, and binds the cascade. Soft-ref mol prove remains constructive existence **in software**. Soft-ref paths for Logical Intelligence, World Labs, or OpenIE do not stamp board package measured_j.

§8.5.1 is not an entry in that plateau list. A typed probability, a typed value or not-sure, a legal transition, a low-energy configuration, and an auditable chart are different objects. Hybrid interplay is how Mixture of Limits navigates them. It does not turn any of them into the commit, and it does not turn constraint energy into measured_j.

8.8 Test-time compute as plateau spend

Teach first. Test-time inference (also called **test-time compute**) means spending additional computation *after* training, when answering a query: longer **chain-of-thought** (CoT) traces, search over candidate solutions, **majority vote** across samples, verifier-guided selection, or **process**

reward scoring of intermediate steps. OpenAI's o1 / o3-style systems popularized sequential scaling of reasoning traces at inference (OpenAI, "Learning to reason with LLMs," 2024). Related process supervision trains reward models on step-level labels rather than only final answers (Lightman et al., "Let's Verify Step by Step," arXiv:2305.20050; ICLR 2024). Parallel strategies sample many short traces and aggregate; sequential strategies extend one long trace. Surveys of inference-time scaling for complex tasks document both families and their dependence on verifier quality (e.g. arXiv:2504.00294).

Extra thinking tokens are treated as a new scaling axis parallel to training tokens and parameters. That axis is real as *engineering*: longer verified search can raise accuracy on hard problems when the verifier is good and the budget is spent on the right difficulty band. Empirical work on o1-like models also shows diminishing returns, overthinking, and cases where longer CoTs degrade accuracy or where parallel majority-style methods scale better than unbounded sequential length (e.g. ACL 2025 findings on o1-like test-time scaling; related overthinking analyses).

Mixture of Limits reading. Extra inference tokens are still bits under a budget.

1. **Value of Information (Vol).** Another thinking step is another observation about an internal search state. When the marginal step does not change the decision benefit B, Vol says stop. Majority vote that re-samples without changing B fails the same test.
2. **Grammar coverage.** If Lookup, Formula, or Solver already covers the coordinate, opening a long CoT generator is waste. Covered grammar closes before Model.
3. **Energy / Landauer.** Every additional sampled token and every verifier forward pass erases and rewrites information. Landauer prices irreversible erasure as a thermodynamic lower bound and appears on receipts as a **labeled estimate**. Catalog or OpCounter estimates for test-time paths remain **Estimated / Unmetered** on the soft-ref path (measured_j=None; board_synth_claimed=false).
4. **Certify-before-commit.** A process-reward score or majority winner is a proposal until certify → commitrefuse → receipt. Soft-ref ReplayClass typing still applies: a long reasoning trace does not launder into Deterministic without a certificate path.

Teach-first takeaway. Test-time compute is a plateau spend knob inside the generative corridor. Mixture of Limits navigates whether that budget should open at all, then how little thinking to spend once Vol is zero past the floor. More thinking steps are not free of Vol, grammar, energy, or certify floors.

8.9 Super learning as plateau ensemble spend

Teach first (academic term). **Super learning** (also called the **super learner**) is a cross-validated ensemble / stacking method from targeted learning: fit a library of candidate algorithms, collect out-of-fold predictions, then learn a meta-learner that combines them to minimize cross-validated risk (van der Laan, Polley, and Hubbard, 2007; Polley and van der Laan, "Super Learner In Prediction," U.C. Berkeley Biostatistics Working Paper 266). It is related to stacking as introduced by Wolpert (1992) and adapted by Breiman (1996). A discrete super learner selects the single best candidate by

cross-validated risk; an ensemble super learner learns weights (often non-negative and summing to one) over candidates. The oracle results for cross-validation selectors underwrite asymptotic optimality *relative to the library*, not freedom from information or energy floors.

Label the industry narrative separately. Public “superintelligence” scaling talk (systems that broadly outperform humans across domains) is a *product and aspiration narrative*, not the van der Laan estimator. Academic **super learning** / **super learner** names the cross-validated stack; it is not a product brand and is not marketing AGI.

Mixture of Limits reading. Ensembles and stacks are still Model-class residual when they generate; when they combine deterministic candidates they still face Vol and certify limits.

1. **Vol.** Another base learner or another fold is another observation. When the meta-learner cannot change B, stop.
2. **Grammar.** If Formula or Lookup covers the coordinate, stacking neural candidates is excess enumeration under a different name.
3. **Energy labels.** Training and evaluating a library multiplies forward passes. Estimates remain estimates; soft-ref ensembles keep `measured_j=None`; `board_synth_claimed=false`.
4. **Certify.** A stacked prediction is still a proposal until a certificate or typed refuse closes the act. Oracle optimality inside a library does not coerce `ModelGenerated` to `Deterministic`.

Teach-first takeaway. Super learning is cross-validated ensemble selection—a plateau option for combining residual leaves. Mixture of Limits maps it onto floors: better combination on the plateau does not retire Vol, grammar, energy, or certify.

8.10 Live SOTA methods for plateau energy/compute (2025–2026)

Each row is a verified **implementation option for plateau spend**: teach the technique once with a cite, then one Mixture of Limits floor sentence. This is not an uncited acronym dump and not a race table. Author-reported speedups and benchmark scores are **field results**, distinct from OpenIE soft-ref measurements.

Approach	What it is (teach once)	Verified cite	Floor that still binds
Speculative decoding	A small drafter proposes several tokens; the large target model verifies them in parallel, preserving the target distribution while cutting serial decode steps	Leviathan, Kalman, and Matias, “Fast Inference from Transformers via Speculative Decoding,” ICML 2023 (PMLR); Li et al., EAGLE , ICML 2024 (arXiv:2401.15077); see also §9 HCSpec / CAS-Spec	Speeds generation; does not refuse generation when Vol or grammar says stop
Mixture-of-Depths (MoD)	Per-layer top- <i>k</i> routing so only a budgeted subset of tokens enter attention/MLP at each depth; residual skip for	Raposo et al., “Mixture-of-Depths: Dynamically allocating compute in transformer-based language models,” arXiv:2404.02258 (2024)	Adaptive depth <i>inside</i> the generator; not Lookup→Formula→Solver refuse outside generation

Approach	What it is (teach once)	Verified cite	Floor that still binds
	the rest (static compute graph, dynamic token compute)		
Weight / activation quantization	Post-training low-bit weights (and sometimes activations): GPTQ (second-order PTQ), AWQ (activation-aware weight scaling), FP8 / INT8 / INT4 serving formats	Frantar et al., GPTQ, arXiv:2210.17323; Lin et al., AWQ, MLSys 2024 (arXiv:2306.00978); Micikevicius et al., FP8 formats (NVIDIA, 2022)	Cheaper plateau bits per weight; does not invent Vol stop or certify-before-commit
Inference engines (KV systems)	Serving runtimes that page and reuse KV cache: vLLM / PagedAttention; SGLang / RadixAttention for structured LM programs and prefix reuse	Kwon et al., “Efficient Memory Management for Large Language Model Serving with PagedAttention,” arXiv:2309.06180 (vLLM); Zheng et al., SGLang, arXiv:2312.07104	Systems efficiency on the plateau; still serves generators—floors decide whether to open decode
Mixture of experts (MoE)	Sparse gating routes each token to a few expert feed-forward modules so total parameters grow faster than active compute	Shazeer et al., “Outrageously Large Neural Networks,” ICLR 2017 (arXiv:1701.06538); Lepikhin et al., GShard, 2020 (arXiv:2006.16668); Fedus et al., Switch Transformers, JMLR 2022	Capacity <i>inside</i> the generative corridor; Mixture of Limits floors live <i>outside</i> generation (§2, §9)
State-space models (SSMs)	Sequence models with a compact recurrent state and near-linear scaling in context (selective SSM / Mamba) as an alternative or complement to attention	Gu and Dao, “Mamba: Linear-Time Sequence Modeling with Selective State Spaces,” arXiv:2312.00752 (2023); Gu, Goel, and Ré, S4, ICLR 2022	Efficient sequence leaf; still Model LAST when used as generator; Vol and certify unchanged
Retrieval-augmented generation (RAG)	Retrieve documents from an external index, then condition the generator on those passages so knowledge need not live only in weights	Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS 2020 (arXiv:2005.11401)	Retrieval can rhyme with Lookup when the hit is certified; uncertified retrieved text remains proposal, not Deterministic commit
Reasoning models / CoT scaling	Models trained or prompted to emit long intermediate reasoning (o1/o3-style, DeepSeek-R1, QwQ, and cousins)	OpenAI o1 public writeup (2024); DeepSeek-AI et al., DeepSeek-R1 technical report (2025); §8.8	Extra reasoning tokens still hit Vol and energy floors; longer is not always better

Approach	What it is (teach once)	Verified cite	Floor that still binds
Diffusion language models	Discrete or continuous diffusion denoising over tokens or latents rather than purely left-to-right next-token prediction (e.g. score-entropy discrete diffusion)	Lou, Meng, and Ermon, “Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution” (SEDD), ICML 2024 (PMLR v235)	Denoising steps spend bits and joules; Landauer remains a labeled estimate only; certify before commit
Linear attention / retention variants	Attention alternatives with linear or chunkwise cost for long context (e.g. Retentive Network / RetNet multi-scale retention)	Sun et al., “Retentive Network: A Successor to Transformer for Large Language Models,” arXiv:2307.08621 (2023)	Cheaper long context is algorithmic efficiency; floors still stop when Vol or grammar is covered

Teach-first takeaway. Every live SOTA method above optimizes *how* a generator runs or *which* parameters activate on the plateau. Mixture of Limits asks whether the generator should run for the stated benefit, which named floor closes first, and—once Vol is zero past that floor—how to spend least energy/compute on the plateau (Lookup → Formula → Solver → Model LAST).

8.10.1 Adaptive depth, precision, and serving as plateau levers

Three complementary plateau levers sit beside MoE and speculative decode:

1. **Mixture-of-Depths** (Raposo et al., arXiv:2404.02258) learns which tokens need full block compute and which may residual-skip under a fixed FLOP budget. That is generator-internal cheapest-sufficient *depth*—a rhyme with Mixture of Limits cascade, not a substitute for floors outside generation.
2. **Quantization** (GPTQ, AWQ, FP8/INT families) moves fewer bits per weight and raises arithmetic intensity on memory-bound decode. Author-reported speedups and perplexity deltas are field results; soft-ref receipts keep Landauer as a labeled estimate and measured_j=None unless Metered.
3. **Inference engines** (vLLM PagedAttention; SGLang RadixAttention) refuse waste of fragmented KV and repeated prefix compute. That is transmission / memory economics at the serving layer (§8.13), still under Model LAST when the cascade opens the generator.

Floor bind. Adaptive depth, lower precision, and better KV paging optimize plateau spend. They do not retire Vol, grammar coverage, certificate, or settle-refuse. Soft-ref path:
board_synth_claimed=false.

8.11 SSM hybrids, MoE, and implementation efficiency on the plateau

Plateau options (verified methods, not a brand race). Once Model is the residual leaf and Vol is zero past the floor, SOTA at the plateau means spend less energy and move fewer bits for the same certified benefit. The following are implementation options:

- **State-space models (SSMs) / Mamba.** Selective state-space sequence models with linear-time scaling in context (Gu and Dao, 2023; Gu, Goel, and Ré, S4, ICLR 2022). Plateau use: cheaper long-context sequence leaf under Model LAST.
- **SSM + attention + MoE hybrids (Jamba-class).** Interleaved Transformer attention and Mamba layers, plus mixture-of-experts on some MLPs, aimed at high throughput and a smaller key-value cache on long contexts (Lieber et al., “Jamba: A Hybrid Transformer-Mamba Language Model,” arXiv:2403.19887; Jamba-1.5, arXiv:2408.12570). Plateau use: cut memory and raise throughput for long generative contexts. Author-reported benchmark and throughput numbers are **field results**, not soft-ref measured_j.
- **Fine-grained MoE + systems co-design (DeepSeek-class).** DeepSeekMoE (Dai et al., ACL 2024) and DeepSeek-V3 (DeepSeek-AI, arXiv:2412.19437) combine fine-grained MoE, multi-head latent attention, and systems co-design (including reported FP8 mixed-precision training and pipeline overlap). DeepSeek-R1 applies large-scale reinforcement learning for reasoning traces on that efficient base. Plateau use: lower active compute and training/inference cost per benefit. Treat GPU-hour and benchmark figures as **author-reported**, not soft-ref meters.
- **MoE reasoning / agentic releases (GLM-class).** GLM-4.5 and related Zhipu / Z.ai releases (arXiv:2508.06471) use large total parameter counts with smaller active counts per token, plus agentic / reasoning post-training. Plateau use: active-parameter efficiency on available silicon. Scores and active-parameter claims are field/author-reported.

Related plateau levers—quantization, speculative decoding, training-systems overlap—appear again in §8.10 and §8.13. Vendor ranking is out of scope.

Mixture of Limits reading.

1. SSM hybrids and MoE routing are legitimate plateau efficiency leaves. They remain generators. Floors still bind.
2. **Implementation efficiency** (algorithms, precision, routing, RL post-training) lowers device-dollar and joule cost on the plateau. That is applied engineering, not a repeal of Shannon, Howard, or Landauer.
3. No architecture announcement retires Vol, grammar coverage, labeled energy estimates, or certify-before-commit. Soft-ref path: `board_synth_claimed=false`; Jamba / DeepSeek / GLM joules remain field / author-reported.

Teach-first takeaway. SOTA at the plateau is optimize energy/compute with cheapest-sufficient cascade. SSM hybrids, MoE, and systems co-design are how field stacks spend less once the floor is known—not a narrative about which brand won.

8.12 Hardware economics: plateau spend on real devices

State as the study's claim (David Charlot / OpenIE framing), grounded in the sections above.

1. **Computers are hardware.** Every program runs on materials and devices: gates, memory, interconnects, power delivery, cooling. There is no software that escapes physics.
2. **Software is written for hardware.** Instruction sets, memory hierarchies, batch sizes, and quantization schemes exist because of devices. Software is **human applied engineering** under constraints and under economic / survival motivation. It is not a separate rocket-science realm detached from cost, yield, and joules.
3. **Plateau efficiency margins remain.** Inefficient deployment on the newest hardware leaves room. Optimized algorithms, MoE routing, training-systems co-design, and inference tricks on older or cheaper accelerators extract more work per device-dollar on the plateau. DeepSeek / GLM-class implementation efficiency (§8.11) and speculative decoding / quantization stacks (§8.10) are instances. Treat vendor scoreboards as field reports, not OpenIE meters.
4. **Mixture of Limits reading.** Plateau options (SSM hybrids, mixture of experts, test-time compute, diffusion language models, linear attention, retrieval-augmented generation, world models, energy-based hybrids) do not retire floors. The bottleneck remains **applied mathematics embodied in materials and devices**, plus labeled joule accounting. Algorithmic efficiency is real and is exactly why floors matter: once Vol is zero past the floor, cheaper implementations are the remaining work, and they still stop when Vol, grammar, or energy says stop. Estimates (estimated_j, Landauer as labeled estimate) are not board package energy. Soft-ref path: measured_j=None; board_synth_claimed=false.

Edge / TinyML floors (verified cites). Coin-cell and MCU closes are not tiny transformers by default. MCUNet co-designs TinyNAS + TinyEngine for ImageNet-class inference on microcontrollers (Lin et al., NeurIPS 2020, arXiv:2007.10319). T-MAC accelerates low-bit LLM matmul on edge CPUs via table lookup without dequantization (Wei et al., arXiv:2407.00088). Mixture of Limits reading: Formula+LUT refuse under device SRAM/Flash is the lawful edge close; a quantized residual Model leaf is still Model LAST.

Neuromorphic / thermodynamic materials options (field / soft-ref). Loihi-2 neuromorphic mapping of a MatMul-free LLM reports author-estimated throughput and mJ/token vs edge-GPU transformers (Abreu et al., arXiv:2503.18002)—field silicon results, not soft-ref measured_j. Digital thermodynamic-computer proposals for generative sampling (e.g. CN101, arXiv:2608.00754) and company thermodynamic platforms rhyme conceptually with soft-ref ThermoSettle DeviceKind; package joules stay soft-ref until Metered. Bennett reversible computing remains settled theory (§8.0); product reversible ASICs are out of soft-ref prove scope.

Bridge. §8.0.1 already stated that awareness of floors is not the gap; embodiment is. §8.12 adds the economic corollary: markets reward plateau spend that extracts more work per device-dollar, and Mixture of Limits is the navigation law that binds that spend to floors when more work buys nothing. §8.13 surveys public transmission / gearing / cascade / memory-context / test-time techniques as plateau options—still generators until refuse is law.

8.12.1 Hardware-paired navigation (2025–2026 peers)

Mixture of Limits is not a software-only order. Navigation is Lookup → Formula → Solver → Model LAST **on hardware that is available, accessible, and capable** ([Metabolic Intelligence](#) states the three tests). Hooker’s hardware lottery remains settled law: ideas look like progress when they fit installed chips and kernels (Hooker, 2021; [doi:10.1145/3467017](#)).

Recent field work strengthens the pairing without replacing the navigation law. Huang et al. (arXiv:2507.19142, Jul 2025), **A3D-MoE**, co-design MoE routing with 3D heterogeneous integration, score-aware HBM access, and attention–MoE fusion; author-reported energy cuts of about 2×–4× are **their** architecture result, not OpenIE measured_j. Chung, Wu, Ma, and Chowdhury (arXiv:2601.22076, 2026), **Where Do the Joules Go?**, measure 1,858 inference configurations on H100/B200 and map energy to latent factors across algorithm, software, and hardware layers—task type alone can swing LLM energy per response by about 25×. That is plateau diagnosis of spend on available fabric. It is not Lookup → Formula → Solver refuse.

Cross-links. [Notational Intelligence](#) owns certify-before-commit on the irreversible branch. [Satiation](#) owns the economic stop when completeness holds or scarce joules bind. [Metabolic Intelligence](#) owns the envelope and fabric interface tax. Soft-ref path: estimates ≠ board package energy; board_synth_claimed=false.

8.13 Plateau techniques: transmission, gearing, cascade, memory/context, test-time

Thesis of this section (SOTA at the plateau). Public research programs optimize **transmission** (how bits move through attention, KV cache, and interconnect), **gearing / MoE routing** (which expert / path / attention mode fires), **cascade** (early-exit prefill, self-decoder then cross-decoder; Lookup→Formula→Solver→Model LAST as OpenIE order), **memory/context** (long-horizon KV, conditional memory, hybrid attention windows), and **test-time** (reasoning budgets, multi-token prediction, speculative decode) under hardware and economic constraints. That is **plateau spend**: algorithmic efficiency on available silicon once Model is open. These programs approach Mixture of Limits ideas—cheapest sufficient compute, refuse waste of HBM / SSD / prefill FLOPs—without naming the law. Unbounded parameter counts and optimization still dominate until memory/context and refuse are first-class. Hardware economics (§8.12) applies: author-reported FLOPs, cache bytes, and API prices are **field / author-reported**, not OpenIE soft-ref measured_j. Estimates ≠ board package joules. Labs are cited as sources for plateau options by method.

Teach the shared primitives before the technique sections:

Primitive	Meaning in this survey	Floor bind
YOCO (You Only Cache Once)	Decoder–decoder layout: a self-decoder builds shared global KV once; a cross-decoder reuses it via cross-attention; prefill can early-exit after the self-decoder (Sun et al., NeurIPS 2024; arXiv:2405.05254)	Cuts KV memory and prefill work; still a generator architecture—Vol / grammar / certify floors remain
HySparse2		

Primitive	Meaning in this survey	Floor bind
	Hybrid sparse attention with two-level KV sharing: outer KV Bridging (YOCO-style self-/cross-decoder, full-attention layers only) plus inner KV Reuse (sparse layers reuse full-attention KV and token-level top-k); prefill exits after the self-decoder (Wei, Gao, ..., Luo / Xiaomi LLM-Core; arXiv:2609.26368)	Compresses long-context transmission cost; does not invent a refuse floor outside generation
CSA2 / CED	DeepSeek-V4.1-Flash: Causal Encoder–Decoder (CED, YOCO-inspired) plus Compressed Sparse Attention 2 (Full / Reindex / Reuse modes, hierarchical sparse indexer, FP4 main KV)	Extreme KV and prefill compression for agents; still MoE generation
MoE routing	Gate selects a small active expert set per token (DeepSeekMoE, Qwen MoE, Kimi, Sarvam, Solar, ...)	Gearing inside the generative corridor—cheaper than dense, not Mixture of Limits floors
MTP / speculative path	Multi-token prediction or draft modules densify training and accelerate decode	Test-time / decode gearing; energy and Vol floors still bind
Cache-hit pricing	API bills cache-hit input far below cache-miss; long sessions that break prefix reuse jump cost	Hardware-economics transmission price; field meter, not soft-ref measured_j

8.13.1 Transmission: attention, KV cache, and interconnect cost

YOCO → CED / CSA2 (DeepSeek-V4.1-Flash). YOCO showed that caching global KV once and early-exiting prefill after a self-decoder cuts memory and latency by orders of magnitude on long contexts (Sun et al., 2024; arXiv:2405.05254). DeepSeek-V4.1-Flash pushes that lineage into agent workloads: a multimodal MoE with 552B backbone parameters, up to **1M** context, Causal Encoder–Decoder (CED) activating **8B** parameters per token in prefill and **16B** in decode, Compressed Sparse Attention 2 (CSA2) with Full / Reindex / Reuse modes, FP4 main KV, SWA Bounded Replay, Engram conditional memory, and DSpark speculative decoding (DeepSeek-AI, arXiv:2609.19969; weights at huggingface.co/deepseek-ai/DeepSeek-V4.1-Flash). Author-reported global KV footprint is **890 bytes/token** (~1/4 of V4-Flash); persistent KV with SWA Bounded Replay ~1/8 of V4-Flash. That is **transmission** law under HBM/SSD constraints—not a board joule claim.

HySparse2 (Xiaomi LLM-Core). HySparse2 (Wei, Gao, ..., Cao, Luo / Xiaomi LLM-Core, arXiv:2609.26368) teaches two-level KV sharing for long-horizon agents: KV Bridging (YOCO-style) so prefill can exit after the self-decoder, plus KV Reuse with **token-level** sparse selection and a forced local window (no separate SWA branch in sparse layers). On matched 80B-A3B MoE ablations, author-reported 1M-token prefill FLOPs drop ~2.92× vs HySparse and ~5.02× vs Hybrid SWA; KV cache ~2.69 GB vs 6.72 / 12.09 GB. Transmission compression under device memory—still a generator stack.

Pricing / cache break (hardware economics). DeepSeek’s public API prices deepseek-flash (served as DeepSeek-V4.1-Flash) with a large gap between cache-hit and cache-miss input (e.g. peak cache-hit \$0.006 vs cache-miss \$0.3 per 1M input tokens; api-docs.deepseek.com/quick_start/pricing). Field discussion on Hacker News notes sessions that lose prefix cache around ~60% of a context window and suddenly pay ~50x more (news.ycombinator.com/item?id=49735410)—an operator anecdote, not an OpenIE meter. Mixture of Limits bind: cheap cache reuse is cheapest-sufficient transmission; when reuse breaks, escalated spend is still generative corridor cost, not a certified refuse.

Interconnect locality (Huawei LocMoE / PanGu-Σ). LocMoE reduces MoE communication overhead on Ascend clusters (IJCAI 2024; arXiv:2401.13920). PanGu-Σ targets sparse trillion-scale sparse heterogeneous compute (arXiv:2303.10845). **Floor bind:** routing locality as transmission cost on available accelerators—hardware economics, not soft-ref joules.

8.13.2 Gearing / MoE routing: which expert or path fires

DeepSeekMoE / V3. DeepSeek-V3 is a Mixture-of-Experts (MoE) language model with 671B total parameters and 37B activated per token, using Multi-head Latent Attention (MLA), DeepSeekMoE routing, auxiliary-loss-free load balancing, and a Multi-Token Prediction (MTP) training objective that can also support speculative decoding (DeepSeek-AI, arXiv:2412.19437; earlier DeepSeekMoE at ACL 2024). That is **gearing**: only a sparse expert path runs per token. It approaches cheapest-sufficient *inside* generation.

Xiaomi MiMo (hybrid attention + MoE). Spelling: Xiaomi (not “Xiami”). Public LLM line: **MiMo**. MiMo-7B is a dense reasoning model pretrained on ~25T tokens with MTP and RL post-training (Xiaomi LLM-Core, arXiv:2505.07608). MiMo-V2-Flash is a 309B-total / 15B-active MoE with hybrid Sliding-Window Attention (SWA) and global attention (5:1, 128-token window), MTP reused as speculative draft, native context extended to 256K (arXiv:2601.02780). Public reporting attributes MiMo V3’s intended architecture to HySparse2 under Xiaomi MiMo lead Fuli Luo (team materials / TechNode 2026-09-24; HySparse2 paper authorship). Treat V3 product status as field announcement until a full V3 technical report ships; the **routing / cascade / transmission** claims bound to the HySparse2 paper. **Floor bind:** Xiaomi compresses long-context transmission and prefill cascade under device memory—Mixture of Limits—adjacent cheapest-sufficient—while remaining MoE/hybrid-generator first.

Qwen / Alibaba. Qwen3 MoE uses fine-grained expert segmentation (e.g. 128 experts, 8 active; no shared experts in the Qwen3 MoE design) plus a unified thinking / non-thinking mode and thinking-budget control (Qwen Team, arXiv:2505.09388). Qwen3-Next further sparsifies (public Alibaba materials: hybrid attention, highly sparse MoE, MTP). **Floor bind:** gearing + test-time budget—still generative; budget is not Vol refuse outside the model.

GLM / Zhipu. Already in §8.11: GLM-4.5 MoE reasoning / agentic releases (arXiv:2508.06471). **Floor bind:** active-parameter efficiency on silicon; floors outside generation remain Mixture of Limits law.

Moonshot / Kimi. Kimi K2 is a ~1.04T-total / ~32B-active MoE with MLA-class attention and MuonClip training (MoonshotAI, arXiv:2507.20534). **Floor bind:** extreme sparse gearing; agentic length still hits energy/Vol.

Sarvam. Sarvam-30B is a public MoE (~30B total / ~2.4B active; 128 sparse experts + shared expert; top-6 routing; GQA) aimed at multilingual / Indic-language reasoning with efficient deployment (Sarvam model card / docs; Hugging Face `sarvamai/sarvam-30b` config). **Floor bind:** expert gearing for multilingual benefit under smaller active FLOPs.

Other sourced MoE / efficiency lines (routing-relevant only).

- **Baichuan.** Baichuan 2 public report is dense 7B/13B training systems work (arXiv:2309.10305)—include only as efficiency-under-hardware, not as MoE routing. **Floor bind:** systems gearing ≠ Mixture of Limits cascade.
- **ByteDance Seed.** Public Seed / Doubao lines emphasize long-context and agent products. Where MoE or sparse attention is documented, bind as generative gearing under §8.12 economics.
- **SB Intuitions (SoftBank group).** Sarashina2-8×70B is a Japanese MoE upcycled from Sarashina2-70B (8 experts, top-2 active; public SB Intuitions release 2024). **Floor bind:** sparse gearing under language-local silicon economics.
- **Upstage.** Solar Open / Solar Open 2 public MoE lines (e.g. ~100B-class with ~12B active; later ~250B / ~15B active with hundreds of routed experts; arXiv:2601.07022 and Upstage blogs). **Floor bind:** MoE gearing on constrained GPU budgets.
- **Krutrīm.** Public Krutrīm LLM technical report describes a dense multilingual foundational model (arXiv:2502.09642). It is **not** an MoE routing paper.
- **AI4Bharat.** Strong open Indic datasets and benchmarks (e.g. IndicXTREME / related suites) that other systems evaluate against—infrastructure for grammar coverage of Indic languages, not a MoE router itself.

8.13.3 Cascade: early-exit prefill and staged decode paths

Cascade here means staged work: finish a cheaper stage, then optionally open a costlier one—self-decoder then cross-decoder; prefill then decode; or OpenIE Lookup→Formula→Solver→Model LAST.

- **YOCO / CED early-exit.** Prefill can exit after the self-decoder once global KV is built (Sun et al., 2024; DeepSeek-V4.1-Flash CED). That is a generator-internal cascade that refuses *waste of prefill FLOPs*, not a Vol refuse outside generation.
- **HySparse2 prefill exit.** Same YOCO-style exit after KV Bridging; author-reported 1M-token prefill FLOPs compressions above (§8.13.1).
- **CED active-parameter schedule.** DeepSeek-V4.1-Flash activates **8B** per token in prefill and **16B** in decode—cascade of capacity with stage, still MoE generation.
- **OpenIE contrast.** Mixture of Limits cascade is Lookup → Formula → Solver/settle → Model LAST with typed refuse. Field cascades above densify or sparsify *generation*; they do not certify commit/refuse before irreversible effect.

8.13.4 Memory / context: long-horizon KV and conditional memory

- **DeepSeek-V4.1-Flash.** 1M context; FP4 main KV; **890 bytes/token** global KV; SWA Bounded Replay; Engram conditional memory (arXiv:2609.19969). Memory/context is becoming first-class; settle-refuse and Vol outside generation are not.
- **Xiaomi MiMo / HySparse2.** MiMo-V2-Flash native context to 256K with hybrid SWA/global attention; HySparse2 author-reported KV cache ~2.69 GB at 1M tokens on the matched MoE ablation (vs 6.72 / 12.09 GB baselines).
- **NAVER HyperCLOVA.** HyperCLOVA X THINK technical materials emphasize reasoning / long context (arXiv:2506.22403)—include as context / test-time path, not as MoE gearing unless a MoE card is cited.
- **Floor bind.** Compressing KV and extending context refuses waste of HBM/SSD under silicon constraints—Mixture of Limits—adjacent on transmission—while unbounded MoE capacity and optimization still dominate until refuse is law.

8.13.5 Test-time: reasoning budgets and speculative paths

DeepSeek-R1 (test-time reasoning path). DeepSeek-R1 incentivizes reasoning via large-scale reinforcement learning (RL), with R1-Zero showing emergent reflection/verification behaviors and R1 adding cold-start data plus multi-stage training; public report arXiv:2501.12948, with a Nature paper (Guo et al., 2025, DOI [10.1038/s41586-025-09422-z](https://doi.org/10.1038/s41586-025-09422-z)). R1 dynamically allocates more reasoning tokens to harder problems—a test-time cascade—while still overthinking on easy items. Mixture of Limits reading: longer chains are not free of Vol or energy floors (§8.8).

MTP / speculative decode. DeepSeek-V3 MTP and DeepSeek-V4.1-Flash DSpark; MiMo MTP reused as speculative draft; Qwen3-Next MTP—decode gearing inside the generative corridor.

Qwen thinking budget. Unified thinking / non-thinking mode with thinking-budget control (arXiv:2505.09388)—test-time spend knob, not Vol refuse outside the model.

HyperCLOVA X THINK. Reasoning / long-context materials (arXiv:2506.22403)—test-time / context path as above.

Floor bind (DeepSeek aggregate). DeepSeek repeatedly refuses *waste of KV / prefill / active parameters*, yet still scales unbounded MoE capacity and optimization. Memory/context is becoming first-class; settle-refuse and Vol outside generation are not.

Section close. Across these sources, the convergent plateau move is: move fewer bits (YOCO / CED / HySparse2 / FP4 KV), fire fewer experts (MoE), and spend test-time tokens where difficulty rises (R1 / thinking budgets)—SOTA energy/compute options on available silicon. That converges toward Mixture of Limits **floors** (stop waste; cheapest sufficient Lookup → Formula → Solver → Model LAST) without stating Vol, grammar, certificate, or settle-refuse as law. Unbounded params + optimization still dominate until memory/context *and* refuse are first-class. Hardware economics (§8.12) is the correct reading of API cache pricing and author-reported KV bytes: field reports, not measured_j.

8.14 Race to the plateau floors across AI/ML fields

Thesis. The industry narrative frames AI progress as unbounded scale—more parameters, more tokens, more experts. Mixture of Limits reframes that race: **it is a race to the true plateaued floors.** Across every major AI/ML field below, the plateau move is real engineering (cheaper bits, fewer steps, better uncertainty, better routing). The floor that binds is still Vol, grammar coverage, labeled energy, certificate, or settle-refuse. Once Vol is zero past that floor, optimize energy/compute there—Lookup → Formula → Solver → Model LAST—rather than escalate generation.

Depth over encyclopedia: one verified cite per row; author-reported speedups and scores are **field results**, not soft-ref measured_j. Soft-ref path: board_synth_claimed=false.

Field	Plateau move	Floor that binds	Cite
RL / RLHF / process reward	Preference reward models + PPO (or process reward on steps) reshape the generative policy toward preferred traces	Reward shaping is still proposal until certify; extra RL / CoT tokens hit Vol and energy once B is covered (§8.8)	Ouyang et al., InstructGPT / RLHF, NeurIPS 2022, arXiv:2203.02155; Lightman et al., process supervision, arXiv:2305.20050
Diffusion (image/video) + distillation	Latent diffusion + step distillation cut denoising compute while preserving sample quality	Denoising steps spend bits/joules; Landauer stays a labeled estimate; certify before commit of generated media	Rombach et al., Latent Diffusion, CVPR 2022; Salimans & Ho, Progressive Distillation, ICLR 2022, arXiv:2202.00512; Blattmann et al., Stable Video Diffusion, arXiv:2311.15127
Graph neural nets	Message-passing GCNs / cousins compress relational structure into local updates	Graph message passes remain Model-class when generative; structure that is Lookup/Formula-covered closes before GNN	Kipf & Welling, GCN, ICLR 2017, arXiv:1609.02907
Multimodal fusion	Contrastive image–text alignment (CLIP-class) shares a joint embedding for zero-shot transfer	Aligned latents do not invent Vol stop; multimodal generation still Model LAST under certify	Radford et al., CLIP, ICML 2021 (PMLR v139)
Federated learning	FedAvg aggregates client updates without centralizing raw data; communication rounds become the scarce resource	Comm rounds are plateau cost; Vol still prices whether another round changes B; energy labels stay Estimated unless Metered	McMahan et al., FedAvg, AISTATS 2017 (PMLR v54)

Field	Plateau move	Floor that binds	Cite
Continual learning	Elastic Weight Consolidation (and cousins) protect important weights against catastrophic forgetting across tasks	Retaining old tasks is plateau memory economics; does not retire grammar coverage or settle-refuse on the new coordinate	Kirkpatrick et al., EWC, <i>PNAS</i> 2017. DOI: 10.1073/pnas.1611835114
Pruning / sparsity (beyond MoD)	Lottery-ticket iterative magnitude pruning finds sparse trainable subnetworks that match dense accuracy	Sparse generators still generate; floors outside generation decide whether to open decode (§8.10 MoD for token-depth sparsity)	Frankle & Carbin, Lottery Ticket, ICLR 2019, arXiv:1803.03635
Knowledge distillation	Teacher→student soft-target transfer compresses ensemble or large-model behavior into a cheaper deployable student	Distilled student is cheaper plateau leaf; Vol/grammar still stop escalation when covered	Hinton, Vinyals & Dean, Distilling the Knowledge in a Neural Network, arXiv:1503.02531 (2015)
NAS / hardware-aware NAS	Search (RL or supernet) for architectures; Once-for-All specializes one trained net to device latency without retraining	Architecture search optimizes plateau spend on silicon; does not name refuse outside generation	Zoph & Le, NAS with RL, ICLR 2017, arXiv:1611.01578; Cai et al., Once-for-All, ICLR 2020, arXiv:1908.09791
Bayesian deep learning / uncertainty	MC dropout (and variational cousins) approximate posterior predictive uncertainty at test time	Uncertainty scores are proposals for refuse thresholds—not automatic Mixture of Limits receipts until typed certify	Gal & Ghahramani, Dropout as Bayesian Approximation, ICML 2016 (PMLR v48), arXiv:1506.02142
Conformal prediction	Distribution-free prediction sets with finite-sample coverage under exchangeability; abstention when the set is too large	Closest formal cousin to typed refuse: coverage-calibrated abstain rhymes with settle-refuse / Vol stop	Angelopoulos & Bates, Conformal Prediction: A Gentle Introduction, arXiv:2107.07511; Vovk, Gammerman & Shafer, <i>Algorithmic Learning in a Random World</i> (2nd ed., 2022)
Causal ML	Structural causal models / do-calculus separate intervention from association	Causal identification is Formula/Solver grammar when the graph is known; associative generators stay Model LAST	Pearl, <i>Causality</i> (2nd ed.); Pearl, Causal inference in statistics: An

Field	Plateau move	Floor that binds	Cite
			overview, <i>Stat. Surv.</i> 2009
Active learning (Vol cousin)	Query the label that most reduces expected risk / uncertainty under a labeling budget	Settles-class active selection is Howard Vol operationalized for labels; stop when marginal label Vol $\leq$ cost	Settles, Active Learning Literature Survey, U. Wisconsin TR 1648 (2009); Howard (1966) Vol (§7.1)
Compression / distillation cascades	Progressive or multi-stage distill (diffusion steps; teacher→student chains) compound plateau compression	Cascaded distill remains generative efficiency; OpenIE cascade is Lookup→Formula→Solver→Model LAST with typed refuse	Salimans & Ho (2022); Hinton et al. (2015); §8.13.3 field early-exit cascades
CUDA Graphs / kernel fusion	Capture static GPU work as a replayable graph; fuse kernels to cut launch and memory traffic	Systems plateau spend on the device; still serves generators—floors decide whether work runs	NVIDIA CUDA Programming Guide (CUDA Graphs); PyTorch CUDA Graphs / <code>torch.compile</code> reduce-overhead notes
Agent memory systems	OS-inspired paging of agent context (main vs external memory; MemGPT-class) under fixed context windows	Memory hierarchy is transmission/context economics (§8.13.4); unbounded agent loops still hit Vol/energy; certify before irreversible tool effects	Packer et al., MemGPT, arXiv:2310.08560 (2023)
RAG variants (GraphRAG)	Build entity/community graph summaries; map-reduce over communities for global corpus questions	Graph index can rhyme with Lookup when certified; uncertified community text remains proposal (§8.10 RAG row)	Edge et al., GraphRAG, arXiv:2404.16130 (2024)
Structured generation (beyond GCD/CRANE)	FSM / regex / CFG-guided decoding (Outlines-class) masks illegal tokens at sample time	Constrains tokens ; Mixture of Limits constrains commit (§9 GCD/CRANE). Grammar coverage can close at Formula without an LLM	Willard & Louf, Efficient Guided Generation for LLMs (Outlines), arXiv:2307.09702 (2023)

Teach-first takeaway. Every row races toward a cheaper or safer plateau. None retires Mixture of Limits floors. The true race across AI/ML is to **find the floor, then optimize energy/compute on it**—not to pretend scale has no stop.

Deliberate gaps (not missing awareness). Exhaustive RL algorithm taxonomies, product-only video systems without public methods papers, vendor CUDA Graph marketing slides without a public programming-model cite, and geography-as-organizing-axis surveys stay out: depth on floors beats name-dumps. Formal verification / LeanDojo-class theorem proving already sits in the industry-gap reference block as a Solver cousin, not repeated here.

9. Mixture of Limits is not MoE, and the model-to-model SOTA table

9.0 Spine (history + mathematics; SOTA at the plateau)

Mixture of Limits is proven as an **information-theoretic** approach rooted in **physics**: formula/law discovery in the Newton-Kepler lineage, not excess generation. Academia already has Shannon → Landauer → complexity/Vol → symbolic and energy floors; Mixture of Limits **embodies** those floors in a commitrefuse cascade. The remaining bottleneck is applied math × materials (hardware), not an awareness gap. The industry race is a **race to the plateau floors**, not unbounded scale. The table below situates field systems so readers see what they optimize (**which neural generator runs**). For latent depth, Logical Intelligence EBM hybrids, World Labs world models, and **SOTA at the plateau** (§8.7)—test-time, super learning, live SOTA methods, SSM/MoE/implementation efficiency, hardware economics, transmission/gearing/cascade/memory-context/test-time techniques, and the cross-field floors table (§8.14)—see §8.4-§8.14: once Vol is zero past the floor, those methods are plateau spend options under Lookup → Formula → Solver → Model LAST. Mixture of Limits asks whether a generator should run at all, then how little energy to spend on the plateau, converging to Vol/floors the way physics converged to predictive laws.

9.1 Law contrast (not a rebrand)

Mixture of Experts (MoE) places experts **inside** a generative corridor: a gate selects parameters; success is next-token likelihood; capacity grows by adding experts to the same act. Mixture of Limits places named **floors outside** generation: Lookup → Formula → Solver/settle → **Model LAST**; success is closed grammar + certified commitrefuse + joule receipt; the model is a demoted residual leaf, cold by default (`allow_model=false`).

Axis	MoE / model↔model field	Mixture of Limits
Where the mixture lives	Inside the model (experts, draft/target pairs, strong/weak LLMs)	Outside generation (named floors + cascade gears)
Routing object	Which model (or draft head) answers	Which tier closes: Lookup / Formula / Solver / Model
Refuse	Rare; usually escalate to a larger model	First-class: Vol, grammar, settle, certificate, primitive_gap, fabric_unavailable, ...
Energy labels	Often latency/cost proxies; some papers meter GPU energy for LLM serving	Receipts: estimated_j labeled; soft-ref measured_j=None; board_synth_claimed=false
		Certified commit + typed ReplayClass + receipt

Axis	MoE / model↔model field	Mixture of Limits
Success metric	Tokens/s, cost@quality, speculative speedup	

RouteLLM, GreenServ, and HCSpec optimize **which neural generator runs** (cost, Wh, or decode speedup inside the generative corridor). Mixture of Limits asks whether a generator should run at all.

9.2 Field SOTA (all model↔model routing / speculation / constrained generation)

Every row below is **model↔model**: routers choose among LLMs; speculative methods draft-then-verify with draft/target models; constrained decoding still samples from an LLM under a grammar. None implements Lookup→Formula→Solver→Model LAST with refuse-to-commit as the close law.

Measurement labels: **Metered** | **Estimated** | **Unmetered**; use a paper's own method section for its label. Field joules stay field results. Soft-ref prove path: **Unmetered** (software-ref);
board_synth_claimed=false.

System	Title (verified)	ID	Class	What it routes / accelerates	Energy / cost as reported by authors	Vs Mixture of Limits
RouteLLM	<i>RouteLLM: Learning to Route LLMs with Preference Data</i> (arXiv); ICLR 2025 venue title uses <i>from Preference Data</i>	arXiv: 2406.18665 · ICLR 2025	Preference router (strong↔weak LLM)	Query → strong or weak LLM	Cost / quality (API \$); not Mixture of Limits Landauer receipts	Model↔model; no Lookup/Formula floor; no typed refuse-to-commit
PEARL (routing)	<i>PEARL: Performance and energy aware routing for LLMs</i>	DOI: 10.1016/j.future.2025.108218 · FGCS 176:108218	Energy-aware multi-LLM router	Query → LLM under energy cap (EMM predicts energy)	Authors report energy-aware routing (GPU/infra); not imported as Mixture of Limits measured_j	Still picks a model; Mixture of Limits may refuse before any model
GreenServ	<i>GreenServ: Energy-Efficient</i>	arXiv: 2601.17551 · ICPE 2026	Contextual bandit multi-LLM router	Query features → LLM; accuracy vs	Metered (authors' GPU Wh via	Model pool routing; Mixture of

System	Title (verified)	ID	Class	What it routes / accelerates	Energy / cost as reported by authors	Vs Mixture of Limits
	<i>Context-Aware Dynamic Routing for Multi-Model LLM Inference</i>			measured GPU energy (Zeus)	Zeus) on their LLM pool (field result, not Mixture of Limits board synth)	Limits cascade is non-neural first
VoltanaLLM	<i>VoltanaLLM: Energy-Efficient and SLO-Aware Disaggregated LLM Serving via Adaptive Frequency Control and State-Space Routing</i>	arXiv: 2509.04827	P/D-disagg serving + frequency + state-space route	Prefill/decode instances + GPU frequency under TTFT/ITL SLOs	Metered (authors; pyNVML on A100/GH200) up to ~36.3% E2E GPU energy vs max-freq baseline (field serving result)	Routes instances of the same generative stack ; Mixture of Limits may never open decode
HCSpec	<i>HCSpec: Two-Tier Horizontal Cascade Speculative Decoding for High-Efficiency Large Language Model Inference</i>	DOI: 10.18653/v1/2026.acl-long.353 · ACL 2026	Speculative decoding (draft↔target)	Position-specialized draft cascade → target verify	Latency speedup (vs EAGLE-3 / AR); energy not Mixture of Limits claim	Speeds generation; does not refuse generation on floors
CAS-Spec	<i>CAS-Spec: Cascade Adaptive Self-Speculative Decoding for On-the-Fly Lossless Inference Acceleration of LLMs</i>	arXiv: 2510.26843 · NeurIPS 2025	Self-speculative cascade (DSIA drafts)	Draft stages from target (sparsity/quant) + DyTC	Latency speedup ~1.1x-2.3x AR (authors); lossless tokens	Model-internal draft hierarchy ≠ Mixture of Limits floors
GCD						

System	Title (verified)	ID	Class	What it routes / accelerates	Energy / cost as reported by authors	Vs Mixture of Limits
	<i>Grammar-Constrained Decoding for Structured NLP Tasks without Finetuning</i>	arXiv: 2305.13971 · DOI: 10.18653/v1/2023.emnlp-main.674 · EMNLP 2023	Grammar-constrained LLM decoding	Mask logits to CFG	Unmetered / quality metrics	Constrains tokens ; Mixture of Limits constrains commit
CRANE	<i>CRANE: Reasoning with constrained LLM generation</i>	arXiv: 2502.09061 · ICML 2025 (PMLR v267)	Reasoning-augmented constrained decoding	Alternate unconstrained reason ↔ constrained answer	Accuracy on GSM-symbolic / FOLIO; not Mixture of Limits joules	Still LLM generation under grammar; Mixture of Limits can close at Formula without an LLM
Constitutional AI	<i>Constitutional AI: Harmlessness from AI Feedback</i>	arXiv: 2212.08073	Principle-trained refuse / RLAIIF	Critique→revise under a written constitution; RL from AI feedback	Alignment quality / human prefs; not Mixture of Limits joules	Trained textual refuse inside the generator; Mixture of Limits typed certify-before-commit + receipted floors outside generation
Toolformer / ReAct	Toolformer (NeurIPS 2023); <i>ReAct: Synergizing Reasoning and Acting in Language Models</i>	Toolformer arXiv: 2302.04761 · ReAct arXiv: 2210.03629	Tool-use / agent loop	LM decides when to call APIs / act↔observe	Task accuracy; not Mixture of Limits Landauer receipts	Agent routing among tools still opens a model ; Mixture of Limits may close at Lookup/Formula before any tool or model

Name collision (do not conflate): ICLR 2025 also has *PEARL: Parallel Speculative Decoding with Adaptive Draft Length* (arXiv:2408.11850); speculative, not the FGCS energy router. §9’s PEARL row is the **routing** paper (DOI 10.1016/j.future.2025.108218). Speculative PEARL sits in the same model↔model bucket as HCSpec/CAS-Spec if cited later.

9.3 What Mixture of Limits claims instead

From PLAN prove (see §6):

- 1. **Cascade order is law**, not a heuristic over LLMs: Lookup → Formula → Solver/settle → Model LAST.
- 2. **Refuse-to-commit** is typed and receipted (voi, settle_refuse, efa_certificate, primitive_gap, fabric_unavailable, claim_unknown, ...).
- 3. **Receipt labels** on the proven path: measured_j=None (Unmetered), estimates labeled (**Estimated**), board_synth_claimed=false.
- 4. **ReplayClass** cannot strengthen: ModelGenerated →/Deterministic.
- 5. Soft-ref multi-fabric routing chooses DeviceKind after tier close; still not “pick GPT-4 vs Mixtral.”

9.4 Measurement bounds

Soft-ref / paper-results bound	Allowed
Soft-ref prove path measured_j	Soft-ref prove with measured_j=None (Metered only when a probe returns)
Board synth flag without meter + synth log	board_synth_claimed=false
GreenServ / VoltanaLLM Wh as soft-ref board energy	Cite them as field model-to-model SOTA with their own Metered labels
Tokens/J horse race vs HCSpec	Refuse-before-tokens as the navigation win

Field SOTA above remains **model↔model**. Mixture of Limits contribution is not a better router among generators: it is the navigation law that embodies known information/energy floors, with a prove harness that keeps Model last and refuse first-class.

10. Toward AGI via limits: open questions and companions

If AGI-grade reliability means closing tasks with certificates, receipts, and known floors, not maximizing tokens under an unlimited generative prior, then the historical method already points the way. Kepler and Newton pursued limits: the shortest predictive law that covered the grammar. Shannon priced bits; Howard priced Vol; Landauer and Bennett priced irreversible erasure; Kolmogorov, Solomonoff, and Chaitin priced compression and incompleteness. Recent formula/mechanism systems (AI Feynman, SINDy, PySR, AI-Newton, AlphaGeometry, MWMs) keep proving the same pattern in modern form.

The missing piece is not the idea. Shannon, Howard, Landauer, Bennett, Kolmogorov, Solomonoff, and Chaitin state the floors as settled theory. The bottleneck Mixture of Limits asserts is **applied mathematics embodied in materials and physical hardware**: devices, fabrics, and meters that make cheapest-sufficient close real at the edge, without laundering estimates as board watts, and without treating MoE capacity as a substitute for grammar coverage.

Mixture of Limits says CI should navigate those floors explicitly: Lookup and Formula first, Solver when settle is the right physics, Model last, refuse when Vol or safety says stop. Soft-ref prove shows the navigation law is constructive in software. Companions on this site (NI commit record; Satiation Economic Reality of Satiation) supply interfaces and economic stops. Soft-ref path keeps `board_synth_claimed=false` and `measured_j=None` unless Metered.

Open questions

- Calibrated μ corpora and silicon HW meters (live catalog 258 Present / 258 live gears + HW Gaps soft-ref sims $\times 8$ + Ferric/Stage C inventories are in soft-ref proof; Gap cells retained; `estimates#measured_j`; `stage_c_measured=false` until meters).
- Live WCA MCP / System One pre-gate and leapfrog openie-path ask (adapters remain stubs on the proven path).
- Materials / device paths that make Formula+LUT coin-cell closes dominate generative spend, with Metered probes when package joules are claimed.
- When a Metered `measured_j` path is published beside soft-ref estimates without laundering estimate as board watts.
- How Vol thresholds should be set per domain without smuggling engagement metrics as completeness (see Satiation).
- Whether mechanism libraries (MWM cousins; AI-Newton concept base) should be imported as typed Lookup entries without breaking clean-room prove.
- How to bind third-party EBM settle scores (e.g. Logical Intelligence-style energies) and world-model proposals (e.g. World Labs Atlas/Marble) into certify-before-commit without laundering generative output as Deterministic.
- How to price test-time compute budgets under Vol so adaptive thinking stops when marginal CoT tokens do not change B, without treating longer traces as automatic progress.
- Whether academic super-learner libraries should be imported as typed Solver/ensemble leaves under certify, without equating them to industry superintelligence narratives.
- How to document plateau-efficiency options (DeepSeek / GLM-class MoE and systems co-design; speculative decoding; SSM hybrids; AWQ/GPTQ/FP8; vLLM/SGLang; Mixture-of-Depths) as Estimated field results beside soft-ref receipts, keeping author GPU-hours distinct from Mixture of Limits `measured_j`.
- How typed certify-before-commit should bind **Constitutional AI / RLAIF** textual refuse and **tool-use agent** loops (Toolformer / ReAct) without laundering model-emitted refuse as a Mixture of Limits receipt.
- Whether neuromorphic (Loihi-class) and thermodynamic-compute field stacks should appear as optional DeviceKind under multi-fabric routing once Metered probes exist—without escalating author mJ/token estimates to soft-ref `measured_j`.

- Pause-token / self-paced adaptive compute (theory cousins of Mixture-of-Depths) stay soft-ref until a typed floor maps them; do not invent `measured_j`.
- How to import **conformal abstention** and **Bayesian uncertainty** scores as typed refuse thresholds without laundering model-emitted confidence as a Mixture of Limits receipt.
- Whether **GraphRAG** community summaries and **MemGPT**-class agent memory pages should bind as Lookup when certified, or remain proposal until certify-before-commit.
- How **active-learning Vol** budgets (Settles/Howard) should share threshold machinery with test-time CoT Vol (§8.8) without smuggling engagement metrics as completeness.
- How dual-phase perception (§11.1) should bound Phase-1 transducer energy so typed AST emission stays cheaper than opening Model LAST on unstructured input.
- How Vol / grammar meta-checks (§11.2) stay $O(1)$ Bloom/trie/EBNF relative to the smallest allowed inference leaf.
- How Primitive Distillation (§11.3) certifies Model LAST proposals into new Lookup/Formula entries without laundering uncertified proposals as Deterministic.
- When Tier-2 shunt meters (§11.4) publish package `measured_j` beside Tier-0 analytical and Tier-1 OS telemetry without mixing estimate classes on a receipt.
- How the claim regime, the test-system label, the multi-indicator specification, the metrology programme, the technical report, and the facility reporting duty in §11.4.1 stay off the receipt until a Metered probe returns a joule for the run.

Companions on this site

- [Notational Intelligence as Commit Law](#); commit/refuse record; analytical energy; measurement bounds.
- [Satiation and Scarcity after Free AI](#); Economic Reality of Satiation after free digital inference.
- [Metabolic Intelligence](#); energy-budget embodiment from tag to campus.
- Living figures: [/living/mol/](#) (companions; analytical OpCounter $\neq$ board power; package `measured_j` only when Metered).

Closing sentence. Pursuit of limits (compression into predictive law, priced information, thermodynamic accounting) is the through-line from Kepler and Newton to Shannon, Howard, Landauer, and Kolmogorov. Mixture of Limits is that through-line stated as a navigation law for computer intelligence; embodying it in materials remains the hard problem. Section 11 names the next engineering gaps on that embodiment path. The industry race is a race to the plateau floors, not unbounded scale. Once the floor is known and Vol is zero past it, SOTA at the plateau (§8.4-§8.14) means optimize energy/compute with cheapest-sufficient Lookup → Formula → Solver → Model LAST—latents, EBM hybrids, world models, test-time, ensembles, SSM/MoE efficiency, transmission/gearing/cascade/memory-context techniques, and the cross-field plateau moves of §8.14 as options. They do not retire the floors. Computers are hardware; software is applied engineering under constraints; labeled joule accounting binds.

11. Closing the engineering gaps: dual-phase roadmap

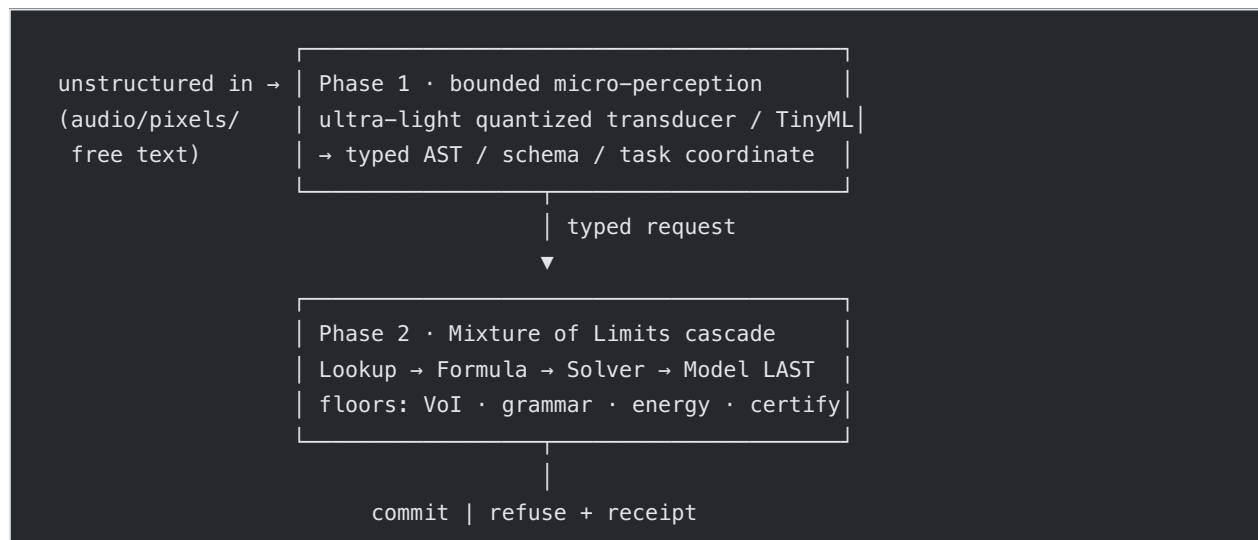
Shannon, Landauer, Howard, and formula-first science already name the floors. Soft-ref mol prove shows constructive existence in software. What remains is engineering: close the gaps that keep Mixture of Limits from owning unstructured front doors, cheap meta-routing, open-ended domains, honest meters, and a usable developer surface—without inventing package joules or demoting the cascade.

This section states those gaps as **plateau work under named floors**. Each item is a gap to close, not a hedge on the law. Estimates remain estimates. `board_synth_claimed=false`. Only Metered paths populate `measured_j`.

11.1 Front-door perception paradox: dual-phase stack

Gap. Mixture of Limits cascade assumes a typed task coordinate: Lookup → Formula → Solver → Model LAST. Real front doors arrive unstructured—speech waveforms, pixels, free text. Without a front gear, the system either refuses everything outside grammar or opens Model LAST as the parser. Both fail the plateau: refuse wastes covered work; parser-as-model burns residual capacity before floors are checked.

Close. Dual-phase Mixture of Limits:



Phase 1 is a **bounded front gear**: fixed vocabulary or schema emitters in the TinyML / quantized-transducer class (MCUNet-class MCU inference; Tiny Transducer-class speech ASR). Clean-room ships a **rule AST transducer** path (`phase1.enabled / mol phase1`) that emits typed ticket/physics/settle asks; unrecognized input refuses parser-as-model. It does not replace Formula. It does not become the substrate. Model LAST remains the residual reasoning leaf inside Phase 2 when grammar and VoI say escalate—not the front-door parser by default.

Plateau reading. Perception spend is plateau spend on the front gear. Once the AST is typed, Vol past the grammar floor is still zero for covered coordinates. Dual-phase does not demote Lookup → Formula → Solver → Model LAST; it feeds that cascade.

11.2 Meta-compute routing overhead

Gap. Vol checks, grammar coverage, and settle-refuse must gate escalation. If those checks cost more than a small inference leaf, the router loses to always-call-model on wall-clock and joules (as estimates). Meta-compute that is heavier than the thing it refuses is not a floor—it is waste on the plateau.

Close. Keep routing **strictly cheaper than the smallest allowed inference**:

Check	Target cost class	Mechanism sketch
Grammar hit / miss	O(1) or O(token) streaming	Trie / Aho–Corasick over typed AST terminals; Bloom filter for negative Lookup
Schema / EBNF accept	linear in input, no model	Deterministic EBNF / PEG parse; reject → refuse or escalate
Vol stop	O(1) threshold table	Precomputed ΔB vs catalog cost; no forward pass
Settle-ready	O(state) local	Ternary / certificate probe before Solver open

```

request AST
  |
  ▼
[Bloom / trie Lookup?] --hit--> Lookup close
  | miss
  ▼
[EBNF / Formula grammar?] --cover--> Formula / Solver
  | uncovered
  ▼
[VoI > 0 AND allow_model?] --yes--> Model LAST → certify
  | no
  ▼
refuse + receipt

```

Rule: if a proposed meta-check needs a neural forward pass, it is not a Mixture of Limits floor check—it is another leaf and must be budgeted as such.

11.3 Static Periodic Stack vs open-ended domains

Gap. Soft-ref navigates a Periodic Stack **subset** (Present/Gap cells; scale notes cite 258 primitives / 33 families). Open-ended domains arrive with undefined grammar. A static stack either silently under-covers or pretends coverage.

Close. Two disciplined exits when grammar is undefined:

1. **Refuse or escalate** with an explicit `primitive_gap` / grammar-miss receipt. Refuse is success when Vol or safety says stop. Escalation opens Model LAST only under budget and `allow_model`.
2. **Primitive Distillation Loop** when a residual is worth keeping:

```
Model LAST proposal
  |
  ▼
certify (typed check / proof / settle)
  | pass
  ▼
AST compile → new Lookup row or Formula identity
  |
  ▼
Periodic Stack Present cell (replay class Deterministic)
```

Uncertified proposals never become Lookup. Distilled entries carry provenance: source receipt, certify method, replay class. This is plateau embodiment of formula-first science: the model proposes; the stack keeps only what certifies.

11.4 Measurement realism: Tier 0 / Tier 1 / Tier 2

Gap. Readers collapse Landauer annotations, OpCounter analytics, OS telemetry, and board shunt readings into one “joules” number. Mixture of Limits forbids that collapse.

Close. Three labeled tiers; only Tier 2 Metered populates package `measured_j`:

Tier	Source	Receipt fields	What it is not
0 Analytical	Catalog μ , OpCounter, Landauer $n k B T \ln 2$	<code>estimated_j</code> , <code>landauer_floor_J</code> , <code>mu_source=catalog</code>	Board package energy
1 OS telemetry	RAPL, NVML, IOReport, powermetrics	Labeled MeasureSource; may set <code>measured_j</code> only when Metered probe succeeds	Shunt-grade package truth; VM/ unavailable stays unset
2 Shunt / package meter	Inline current shunt, certified bench meter	<code>measured_j</code> under Metered; <code>board_synth_claimed</code> only if <code>synth+meter</code> path is actually claimed	Soft-ref default (soft-ref keeps <code>board_synth_claimed=false</code>)

```
Tier 0  estimated_j / Landauer estimate
Tier 1  OS telemetry (RAPL/NVML/...) → measured_j only if Metered label
Tier 2  shunt / package meter      → measured_j (Metered)
```

Measurement facts (unchanged). Soft-ref path: board_synth_claimed=false; measured_j=None unless a Metered probe returns. Estimates ≠ board joules. Never invent measured_j from prove criteria, catalog ratios, or field author GPU-hours.

11.4.1 Public documents, by what they measure

Energy to run is the only true metric of computer intelligence. All other factors collapse to zero. An estimator, a label, a specification, a programme, a technical report, or a facility reporting duty is not a joule on a commit. Mixture of Limits leaves measured_j unset until a Metered probe returns. The documents below are ordered by what each page says the document measures or claims. The critique is the class mechanism: estimator versus a meter. Estimates are not measured joules.

Claim regime. AFNOR SPEC 2314, *Référentiel général pour l'IA frugale*, AFNOR, June 2024. The shop record is 100 pages, with free consultation authorized by AFNOR. The specification states that it was drawn from January 2024 to June 2024, from successive contributions of more than one hundred members of the participating organizations. The ministry press release of 28 June 2024 states 150 contributors and calls the référentiel a first at international level, of voluntary application. The AFNOR news page of 12 July 2024 states a working group of about forty actors and 31 practice sheets. The text contains practice sheets BP 01 through BP 31. Section 4 states what to specify when communicating on the frugal character of an AI service. Annex 1 compares tools that may be used and says the list is not exhaustive. In the column headed measurement of consumption it names CodeCarbon and CarbonTracker, lists direct measurement as an advantage, and lists as a limit that direct measurement is not always possible and that the tools make choices that are not always explicit, for example the emission factor. Practice sheet BP 21 says to record the estimate supplied by those tools (printed there as Code Carbon, Green Algorithms, Carbon Tracker, and MLCO2Impact), then the real consumption of the training. Practice sheet BP 23 is an a priori estimate of model consumption and names CodeCarbon, TensorFlox Profiler, and nvidia-smi as tools that can be used to estimate and to measure. The specification does not require a meter on the commit.

Measured-on-a-test-system label. Umweltzeichen Blauer Engel for ressourcen- und energieeffiziente Softwareprodukte (DE-UZ 215), Umweltbundesamt, edition January 2020, version 6 (July 2026), published by RAL gGmbH. The criteria define the Messsystem as the hardware system on which the product is installed and the measurements are run, including the operating system and its configuration. They also call that system the system under test. For a personal computer or a server, electrical energy demand is determined by integrating electrical power, in watts, over the duration, in seconds, of the standard usage scenario or the long-term test, or by a meter that records electrical work, in watt-seconds, directly. Scenario-test energy is reported net of the electrical baseline, in watt-hours. That is a measurement on a stated test system for a stated usage scenario. It is not the energy of a Mixture of Limits commit.

Estimator variant in the research report beside that label. SoftAWARE, Umweltbundesamt TEXTE 71/2026, completion date May 2023. The report defines two methods. SoftAWARE-Light does without measurements and computes an estimate of energy use and environmental impact from mathematical formulas. The report assigns that variant to cloud and virtualized environments where a physical measurement, or a measurement through server interfaces, is not possible. The full SoftAWARE method does without physical measuring devices and uses existing interfaces of the server manufacturer and the operating system. The indicator Operational Primary Energy Consumption is an estimated server energy use over one hour taken from manufacturer data. The report marks that indicator as no measurement. The report also states that energy from unit tests and integration tests is not a consumer environmental label and does not replace the Blue Angel, because those tests have little to do with later use. A formula estimate is not a measured joule. An interface reading is not the shunt tier.

Multi-indicator specification. ESPECIFICACION UNE 0086:2025, *Medición del consumo energético, huella de carbono, consumo del agua y rendimiento de sistemas de Inteligencia Artificial*, UNE, edition 8 October 2025, corrected 29 July 2026. The UNE catalog page states that the download is free and is sponsored by the Secretaría de Estado de Digitalización e Inteligencia Artificial. The UNE notice of 10 October 2025 states a working group of over 40 members, including certification bodies, and states that the specification defines indicators of direct energy and water consumption during training and inference. The secretariat press of 9 October 2025 states 35 expert entities in its summary line and more than 40 members in the body. That press states a detailed guide for quantifying environmental impact across the life cycle, especially training and implementation, locally and in the cloud. On the pages that state the requirement, the energy number is an indicator of direct energy consumption during training and inference, inside a specification that also covers carbon, water, and performance. Those pages do not name the instrument.

Metrology programme. 《人工智能计量体系和能力建设指引（2026版）》，市场监管总局 和 国家发展改革委, notice of 28 May 2026. The notice states a goal that AI technical performance be measurable, comparable, and traceable. It sets out six parts: foundational support, general technology, core technology, metrology technical specifications, a metrology services industry, and intelligence applied to metrology. It calls for metrology standard devices covering algorithm models, compute efficiency, and data quality, and for work in 14 fields. The notice is a programme for building that capability. It does not require a joule reading of a run. Traceability on this page is a stated goal for technical performance. It is not a reading tied to a measurement standard on a commit.

Technical report. PD CEN/CLC/TR 18145:2025, *Environmentally sustainable Artificial Intelligence*, BSI publication 28 February 2025. A CEN-CENELEC gap analysis dated 18 December 2025 lists the same report, work item JT021010, as published in the JTC 21 column. The published scope describes environmental sustainability issues for those developing or using AI, creates an inventory of impacts and techniques, and states that methods of measuring the environmental sustainability impacts of AI are also quantified. It states that the document is intended to help develop new standards. It states that it is aligned with ISO/IEC TR 20226, whose title on that page is Green and Sustainable AI. A technical report is not a harmonised standard. CEN and CENELEC, in a decision posted 23 October 2025, state that key standards developed under CEN-CLC/JTC 21, requested

under standardization request M/593 and amendment M/613, are to be available by Q4 2026. That page does not name TR 18145 among those drafts. It does not say those drafts are already harmonised standards.

Facility reporting duty. The Agency for Natural Resources and Energy (資源エネルギー庁), explainer of 20 May 2026, states that new measures under the 省エネ・非化石転換法 took effect in April 2026. From the fiscal year 2026 submission, the periodic report adds data-centre electricity use, PUE, エネルギー消費原単位, and future targets. The page defines PUE as the data centre facility's total energy consumption divided by the IT equipment's energy consumption. The same page sets a benchmark of PUE at or below 1.4 by fiscal year 2030 for designated data-centre operators, and a PUE at or below 1.3 for data centres newly built from fiscal year 2029, assessed from the fiscal year after two years of operation. PUE is facility overhead. It is not the energy of a computation.

11.4.2 Research that treats energy as the condition

Public standards have no firm requirement to be energy first and energy bound. Mixture of Limits treats energy to run as the foundation. The Klere energy processing unit is the hardware that can hold a run to that bound. A standard that leaves the meter optional does not. The works below treat energy, power, or a physical bound as a condition of the computation. Energy to run is the only true metric of computer intelligence. All other factors collapse to zero. A number below is that paper's bound or that paper's measurement. It is not package measured_j.

Speed bound from energy. Margolus and Levitin (1998) bound the minimum time for an isolated system to reach an orthogonal state by the average energy above the ground state, $\tau_{\perp} \geq h/(4E)$. Their abstract states that adding one joule of energy cannot raise the processing rate by more than about 3×10^{33} operations per second. Class: physical bound.

Work bound on measurement and erasure. Sagawa and Ueda (2009) derive lower bounds on the work of a measurement and on the work of erasure. The sum is bounded below by the mutual information I gained in the measurement, $W_{\text{meas}} + W_{\text{eras}} \geq k_B T I$. For a symmetric memory the erasure bound matches Landauer. For an asymmetric memory the erasure bound can lie below that match. Class: thermodynamic bound.

Dissipation of nonpredictive memory. Still, Sivak, Bell, and Crooks (2012) set the instantaneous nonpredictive information equal to the average work dissipated when the driving signal changes, in units of $k_B T$. A system built to keep memory, and built to operate at minimum dissipation, has to be predictive. Class: thermodynamic bound.

Measured erasure. Bérut, Arakelyan, Petrosyan, Ciliberto, Dillenschneider, and Lutz (2012) erase one bit stored in a colloidal particle in a modulated double-well trap. They report that the mean dissipated heat saturates at the Landauer bound in the limit of long erasure cycles. Class: measured approach to a thermodynamic bound. That heat is their measurement.

Jun, Gavrilov, and Bechhoefer (2014) erase one bit with a colloidal particle in a feedback trap. Work is computed from the trajectory and the imposed potential. The full-erasure protocol's asymptotic work is compatible with the Landauer bound. The same cycle without phase-space compression is compatible with zero work. Class: measured work of erasure, with a reversible control. That work is their measurement.

Heat of a Turing-machine realization. Kolchinsky and Wolpert (2020) assign a heat function to a physical realization of a Turing machine. For the coin-flipping realization of a universal machine, heat on an input is proportional to the excess length of that input over the shortest program for the same output, up to an additive constant. The minimum heat required to produce any chosen output is bounded by a constant. The expected heat on the coin-flipping input distribution is infinite. Class: thermodynamic bound on a computation. The heat function is not package measured_j.

Energy as the training objective. You, Chung, and Chowdhury (2023) define energy-to-accuracy as average GPU power times time-to-accuracy. GPU power is read with NVML, and the GPU power limit is a training knob. An exploratory run stops when its cost exceeds twice the lowest cost observed so far. The paper reports energy reductions of 15.3% to 75.8% against maximum batch size and maximum power limit, on the GPUs named in the evaluation. Class: energy as the objective of training, with a stop. Those reductions are their NVML measurements.

Energy schedule under a time bound. Chung, Gu, Jang, Meng, Bansal, and Chowdhury (2024) define energy bloat as the energy in a large-model training iteration that can be removed without lengthening the critical path. Perseus profiles the time and energy of each forward and backward computation and sets GPU frequency through NVML. The energy-optimal iteration time is the minimum of the minimum-energy iteration time and the straggler iteration time. The paper reports per-iteration energy reductions of up to 30%, with negligible or no slowdown. Class: energy schedule of a training iteration. Those reductions are their measurements.

11.4.3 Neuromorphic substrates and measured-run energy

Mixture of Limits treats energy to run as the foundation. The works below state energy or power of a computation as a physical quantity on a named platform, or they stop under an energy or accuracy-energy bound. They are ordered by method class, not by region. Energy to run is the only true metric of computer intelligence. All other factors collapse to zero. A number below is that paper's own measurement or that paper's own bound. It is not package measured_j. Estimates are not measured_j.

Physical quantity of a neuromorphic substrate. Merolla, Arthur, Alvarez-Icaza, Cassidy, Sawada, Akopyan, Jackson, Imam, Guo, Nakamura, Brezzo, Vo, Esser, Appuswamy, Taba, Amir, Flickner, Risk, Manohar, and Modha (2014) build TrueNorth: 4096 neurosynaptic cores, 1 million programmable spiking neurons, and 256 million configurable synapses. With 400-by-240 video at 30 frames per second, the chip consumes 63 mW. Class: physical quantity of a neuromorphic substrate. That 63 mW is their measurement.

Frenkel, Lefebvre, Legat, and Bol (2018) build ODIN, a 256-neuron 64k-synapse online-learning digital spiking processor in 28 nm FDSOI. At 0.55 V they report a global energy per synaptic operation of 12.7 pJ at the maximum synaptic-operation rate. With rank-order coding on a single-layer MNIST network after on-chip SDSP learning, they report 15 nJ per inference at 84.5% accuracy. Class: physical quantity of a neuromorphic substrate, and measured energy of a run. Those energies are their measurements.

Measured energy of a run on a named neuromorphic platform. Esser, Merolla, Arthur, Cassidy, Appuswamy, Andreopoulos, Berg, McKinstry, Melano, Barch, di Nolfo, Datta, Amir, Taba, Flickner, and Modha (2016) map deep convolutional networks onto TrueNorth. Across eight vision and speech datasets, single-chip networks run at 1200 to 2600 frames per second and use between 25 and 275 mW. Class: measured energy of a run. Those powers and throughputs are their measurements.

Göltz, Kriener, Baumbach, Billaudelle, Breitwieser, Cramer, Dold, Kungl, Senn, Schemmel, Meier, and Petrovici (2021) train hierarchical time-to-first-spike networks of leaky integrate-and-fire neurons and emulate them on BrainScaleS-2. On downsampled MNIST they report 8.4 μJ per classification at 96.9% test accuracy, with chip power measured at 175 mW during runtime. Class: measured energy of a run. That 8.4 μJ is their measurement.

Blouw, Choo, Hunsberger, and Elias Smith (2019) benchmark a keyword spotter on Loihi against CPU, GPU, Jetson TX1, and Movidius. On Loihi they report 0.00027 joules per inference at 296 inferences per second, with idle 0.029 W and running 0.110 W. Classification accuracy on their test set matches the TensorFlow reference. Class: measured energy of a run. Those joules are their measurements.

Gobieski, Lucia, and Beckmann (2019) run deep neural network inference on an intermittent energy-harvesting MSP430 with Sonic and Tails. From their prototype they take inference energy of about 198 mJ for a tiled baseline and about 26 mJ for Tails. Loop continuation keeps each loop iteration idempotent, so a run continues after a power failure. Tails calibrates its tile size to the energy buffer and halves the tile when a tile does not finish before power fails. Class: measured energy of a run under an energy buffer, with a stop and retry at that buffer. Those millijoules are their prototype figures.

Measured energy of inference on named GPUs. Samsi, Zhao, McDonald, Li, Michaleas, Jones, Bergeron, Kepner, Tiwari, and Gadepally (2023) measure LLaMA inference energy on named NVIDIA V100 and A100 GPUs with nvidia-smi and DCGM. They report energy per second, per decoded token, and per response for LLaMA 7B, 13B, and 65B on Alpaca and GSM8K, including multi-node sharding up to 32 GPUs. For LLaMA 65B they report about 3 to 4 joules per output token. Class: measured energy of a run. Those joules are their measurements.

Luccioni, Jernite, and Strubell (2024) measure energy and carbon for 1,000 inferences across 88 models, 10 tasks, and 30 datasets on NVIDIA A100 GPUs with CodeCarbon. Mean energy per 1,000 queries ranges from 0.002 kWh for text classification to 2.907 kWh for image generation. Multi-purpose generative models emit orders of magnitude more carbon than task-specific models on the same discriminative tasks. Class: task energy read with Code Carbon. Those kilowatt-hours are their measurements.

Energy bound with a stop. Yang, Zhu, and Liu (2019) train deep networks under a quantitative energy budget. The budget is an optimization constraint. Weighted sparse projection and layer input masking enforce it during training. The trained model is designed to meet the budget while maximizing accuracy. Their energy model follows a systolic-array estimate. Class: energy bound on training. That budget is their estimation model. It is not package measured_j.

11.4.4 Computing in the physics: analog, in-memory, and frequency-domain substrates

Mixture of Limits treats energy to run as the foundation. The works below compute with the physics of a device: conductance in a memory array, light through an optical path, the phase of coupled oscillators, the decay of a dynamic memristor. They are ordered by method class, not by region. Each paper reports in its own unit, and the class line names how that number was made. An efficiency ratio stays in with its method named. Energy to run is the only true metric of computer intelligence. All other factors collapse to zero. No number below is package measured_j. Estimates are not measured_j.

Analog in-memory computing on phase-change memory. Ambrogio, Narayanan, Tsai, Shelby, Boybat, di Nolfo, Sidler, Giordano, Bodini, Farinha, Killeen, Cheng, Jaoudi, and Burr (2018) train fully connected networks of up to 204,900 synapses in mixed hardware and software, with long-term weights in phase-change memory and near-linear updates on volatile capacitors. They reach software-equivalent accuracy on MNIST, MNIST-backrand, CIFAR-10, and CIFAR-100, and calculate 28,065 billion operations per second per watt for their implementation. Class: efficiency ratio in operations per second per watt, calculated for the implementation. That figure is their calculation.

Ambrogio, Narayanan, Okazaki, and 21 coauthors (2023) build an analog-AI chip of 35 million phase-change memory devices across 34 tiles that performs matrix-vector multiplication in place. They report up to 12.4 tera-operations per second per watt chip-sustained, software-equivalent accuracy on a keyword-spotting network, and near-software-equivalent accuracy on the MLPPerf RNNT, with more than 45 million weights on more than 140 million devices across five chips. Class: efficiency ratio in TOPS per watt, chip-sustained. That ratio is their figure.

Le Gallo, Khaddam-Aljameh, Stanisavljevic, and 27 coauthors (2023) build a 64-core analog in-memory compute chip in 14 nm CMOS with backend-integrated phase-change memory, digital activation functions, and an on-chip network. For 8-bit matrix-vector multiplication they report a maximum of 16.1 or 63.1 tera-operations per second at 2.48 or 9.76 tera-operations per second per watt, in the four-phase or one-phase read mode, with near-software-equivalent accuracy on ResNet and LSTM networks. Class: efficiency ratio in TOPS per watt, on chip. Those ratios are their figures.

Resistive and memristor crossbars. Wan, Kubendran, Schaefer, Eryilmaz, Zhang, Wu, Deiss, Raina, Qian, Gao, Joshi, Wu, Wong, and Cauwenberghs (2022) build NeuRRAM, a compute-in-memory chip that runs inference inside resistive RAM. They report 1.6-times to 2.3-times lower energy-delay product than prior RRAM compute-in-memory chips across bit precisions, 99.0% on MNIST, 85.7% on CIFAR-10, and a measured 121 fJ per neuron, or 0.95 fJ per weight, for each

noise-injection step of probabilistic sampling. Class: energy-delay product and energy per step, measured on chip. Their expected gain of at least 535 times in energy-delay product from 130 nm to 7 nm is projected from a circuit model.

Yao, Wu, Gao, Tang, Zhang, Zhang, Yang, and Qian (2020) implement a five-layer convolutional network fully in memristor crossbars, eight arrays of 2,048 cells, with hybrid training that absorbs device variation. They report more than 96% accuracy on MNIST and an energy efficiency more than two orders of magnitude greater than state-of-the-art graphics-processing units. Class: efficiency ratio against GPUs. That ratio is their comparison.

Photonic and optical computing. Shen, Harris, Skirlo, Prabhu, Baehr-Jones, Hochberg, Sun, Zhao, Larochelle, Englund, and Soljačić (2017) propose a fully optical neural network and demonstrate its core on a programmable nanophotonic processor of 56 Mach-Zehnder interferometers in a silicon photonic circuit, applied to vowel recognition. Their preprint states a promised gain of at least two orders of magnitude in speed and three orders of magnitude in power efficiency over the state of the art. Class: power-efficiency ratio projected from an architecture model. The demonstration is on chip. The ratio is their projection.

Wang, Ma, Wright, Onodera, Richard, and McMahon (2022) compute optical dot products with up to 505,521 scalar multiplications in one pass of light and run a four-layer MNIST network. They report about 99% accuracy at about 3.1 detected photons per multiplication, and about 90% at about 0.66 photons, about 2.5×10^{-19} J of optical energy per multiplication. At 99% the detected optical energy per inference is about 107 fJ, and about 230 fJ counting the modulator loss. Class: optical energy measured at the detector. It excludes the electronics, and the authors say so.

Chen, Nazhamaiti, Xu, Meng, Zhou, Li, Fan, Wei, Wu, Qiao, Fang, and Dai (2023) build ACCEL, an all-analog chip in which a diffractive optical encoder feeds photocurrents straight into an analog electronic computing chip, with no analog-to-digital converter. They report a systemic energy efficiency of 74.8 peta-operations per second per watt, 4.6 peta-operations per second, 72 ns latency per frame, and 85.5% on Fashion-MNIST. Class: systemic efficiency ratio in operations per second per watt. That ratio is their figure.

Feldmann, Youngblood, Karpov, Gehring, Li, Stappers, Le Gallo, Fu, Lukashchuk, Raja, Liu, Wright, Sebastian, Kippenberg, Pernice, and Bhaskaran (2021) build an integrated photonic tensor core that performs parallel convolutions with phase-change-material memory arrays and soliton microcomb light sources. They report operation at trillions of multiply-accumulate operations per second and a bandwidth exceeding 14 gigahertz. Class: throughput in operations per second, no energy figure given.

Frequency-domain and oscillator computing. Miscuglio, Hu, Li, George, Capanna, Dalir, Bardet, Gupta, and Sorger (2020) compute convolutions as pixelwise multiplications in the Fourier domain. Lenses perform the real-to-Fourier transforms passively at zero static power, and an amplitude mask on about 10^6 display channels multiplies. They report matrices of about 1000×1000 in a single

time step, 100 μ s latency, throughput of peta-operations per second, and 98% on MNIST in a hybrid optical-electronic network. Class: throughput and latency, with zero static power for the transform. The paper gives no energy per operation.

Moy, Ahmed, Chiu, Moy, Sapatnekar, and Kim (2022) build a 1,968-node King's graph array of coupled ring oscillators that settles to the ground state of an Ising problem through phase. They report up to 95% accuracy on randomly generated combinatorial optimization problems, a measured average power of 0.042 W, and fewer than 50 oscillation cycles to resolve. Class: power measured on chip, with settling time in oscillation cycles. That power is their measurement.

Romera, Talatchian, Tsunegi, Abreu Araujo, Cros, Bortolotti, Trastoy, Yakushiji, Fukushima, Kubota, Yuasa, Ernoult, Vodenicarevic, Hirtzlin, Locatelli, Querlioz, and Grollier (2018) train a hardware network of four coupled spin-torque nano-oscillators to recognize seven spoken vowels by tuning their frequencies, with classes read from their synchronization. Their preprint reports recognition rates up to 89% on training data and 88% on testing data, and a cross-validated 84.3% on the testing set. Class: recognition rate on a physical oscillator, no energy figure given.

Torrejon, Riou, Abreu Araujo, Tsunegi, Khalsa, Querlioz, Bortolotti, Cros, Yakushiji, Fukushima, Kubota, Yuasa, Stiles, and Grollier (2017) use a single nanoscale spintronic oscillator, a magnetic tunnel junction, to recognize spoken digits. Their preprint reports recognition rates up to 99.6% with cochlear filtering. Class: recognition rate on a physical oscillator, no energy figure given.

Reservoir computing on a physical substrate. Zhong, Tang, Li, Liang, Liu, Li, Xi, Yao, Hao, Gao, Qian, and Wu (2022) build a fully analog reservoir computer: dynamic memristors form the reservoir and non-volatile memristors form the readout. They report real-time spatiotemporal processing with three orders of magnitude lower power consumption than digital hardware, 96.6% on arrhythmia detection, and 97.9% on dynamic gesture recognition. Class: power ratio against digital hardware. That ratio is their comparison.

11.5 Developer experience roadmap (not shipped product)

Gap. Soft-ref mol prove is a clean-room existence proof. Operators still need a declarative surface, local develop loop, and an honest benchmark story versus MoE routers—without claiming a shipped product that does not exist.

Close. Roadmap only (not claimed shipped):

Declarative mol.yaml sketch

```
# Roadmap sketch – not a shipped product claim
version: 0.1
task:
  id: unit.convert.si.length
  grammar: formula.si_units
cascade:
  allow_model: false
floors: [voi, grammar, energy, efa_certificate]
```

```

phase1:
  enabled: false # typed input assumed; enable for unstructured front doors
measurement:
  tier: 0 # 0 analytical | 1 os_telemetry | 2 shunt
  board_synth_claimed: false

```

CLI loop (shipped in clean-room): `mol prove` (PLAN + product A1–A14) → `mol run --chore` `mol.yaml` → `mol bench` (J/query versus always-model / MoE-sim; **Estimated/Metered** only) → `mol arena` (typed decision / ticket-close / risk head-on versus `frontier_sim` + `system_one_leaf`; metrics `correct_close`, `refuse-when-C=1`, `estimated_j`, `latency`) → `mol phase1` → `mol distill` (Primitive Distillation v1) → `mol dev` (catalog/Stage C/PUBLISH status). Watch/receipt-diff UI polish remains residual. Package joules stay Estimated unless Metered.

Product path. Mixture of Limits as a valued product surface (problem, vs Laya System One, `mol.yaml`, acceptance A1–A14, dual-phase → live in-crate NI certify → satiation architecture, distillation v1, durable episodes, Tier-1/2 meter honesty, Arena head-on) lives under the clean-room tree at `product/` (optional drafting mirror `mixture-of-limits-product`). Canonical code + docs: [openIE-dev/mixture-of-limits](#) (`mol prove` embeds A1–A14; `mol run` / `mol bench` / `mol arena` / `mol phase1` / `mol distill`). Soft-ref constructive existence remains `mol prove`; package joules stay Estimated unless Metered. Compete head-on: `correct_close`, `refuse-when-C=1`, `estimated_j`, `latency` — floors win when they exist; Model LAST when needed; peers age; each stands alone; frontier is expensive/out of road (see `product/COMPETITIVE.md`, `product/ARENA.md`).

Benchmark framing. Compare joules-per-query, correct close, `refuse-when-C=1`, and certify rate of Mixture of Limits cascade against MoE / frontier always-model and System One leaf stubs on the same typed task set (`mol arena`). Report Tier 0/1/2 explicitly. Soft-ref constructive existence is not a board energy score; never invent `measured_j`.

11.6 Roadmap summary

Gap	Close	Ships when
Unstructured front door	Dual-phase: TinyML/transducer → typed AST → cascade	Phase-1 gear + schema emitters under prove
Expensive meta-routing	O(1) Bloom/trie/EBNF/Vol tables	Soft-ref criteria for router cost ≪ leaf cost
Undefined grammar	Refuse/primitive_gap or Primitive Distillation	Certify→AST compile→Lookup/Formula path
Joules collapsed	Tier 0 / 1 / 2 labels; Metered-only <code>measured_j</code>	Receipt schema + optional shunt path
DX / MoE / Arena comparison	<code>mol.yaml</code> , <code>mol prove</code> A1–A14, <code>mol bench</code> , <code>mol arena</code> , <code>mol distill</code> , <code>mol phase1</code> , <code>mol dev</code> status	Shipped in <code>openIE-dev/mixture-of-limits product/</code> ; <code>watch/receipt-diff</code> UI polish residual

Mixture of Limits stays the navigation law. Dual-phase, cheap meta-compute, distillation, measurement tiers, and DX are how the law gets embodied on the plateau—Lookup → Formula → Solver → Model LAST, with perception as a bounded front gear and Model LAST still last.

The documents in §11.4.1 do not set `measured_j`. The works in §11.4.2, §11.4.3, and §11.4.4 do not set it either.

References

Primary Mixture of Limits workspace sources (not peer-reviewed publications): `mixture-of-limits README.md`, `BLUEPRINT.md`, `PLAN.md`; `docs/adjacent-field-hunt.md`; `docs/appendix-mwm-ai-newton.md` (author: David Charlot / OpenIE).

Information theory / decision / thermo / complexity

1. Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423. DOI: [10.1002/j.1538-7305.1948.tb01338.x](https://doi.org/10.1002/j.1538-7305.1948.tb01338.x). Part II: 27(4), 623–656. DOI: [10.1002/j.1538-7305.1948.tb00917.x](https://doi.org/10.1002/j.1538-7305.1948.tb00917.x).
2. Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3), 183–191. DOI: [10.1147/rd.53.0183](https://doi.org/10.1147/rd.53.0183).
3. Bennett, C. H. (1973). Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6), 525–532. DOI: [10.1147/rd.176.0525](https://doi.org/10.1147/rd.176.0525).
4. Kolmogorov, A. N. (1965). Three approaches to the definition of the concept “quantity of information.” *Problemy Peredachi Informatsii*, 1(1), 3–11 (no DOI). English transl. (1968): *International Journal of Computer Mathematics*, 2(1–4), 157–168. DOI: [10.1080/00207166808803030](https://doi.org/10.1080/00207166808803030). See also *IEEE Trans. Inf. Theory* note DOI: [10.1109/TIT.1968.1054210](https://doi.org/10.1109/TIT.1968.1054210).
5. Solomonoff, R. J. (1964). A formal theory of inductive inference, Part I. *Information and Control*, 7(1), 1–22. DOI: [10.1016/S0019-9958\(64\)90223-2](https://doi.org/10.1016/S0019-9958(64)90223-2).
6. Chaitin, G. J. (1977). Algorithmic information theory. *IBM Journal of Research and Development*, 21(4), 350–359. DOI: [10.1147/rd.214.0350](https://doi.org/10.1147/rd.214.0350).
7. Howard, R. A. (1966). Information value theory. *IEEE Transactions on Systems Science and Cybernetics*, 2(1), 22–26. DOI: [10.1109/TSSC.1966.300074](https://doi.org/10.1109/TSSC.1966.300074).
8. Cover, T. M., & Thomas, J. A. *Elements of Information Theory* (2nd ed.). Wiley. Book DOI: [10.1002/047174882X](https://doi.org/10.1002/047174882X).
9. Jaynes, E. T. (2003). *Probability Theory: The Logic of Science* (G. L. Bretthorst, Ed.). Cambridge University Press. DOI: [10.1017/CBO9780511790423](https://doi.org/10.1017/CBO9780511790423).
10. Feynman, R. P. *Feynman Lectures on Computation* (A. J. G. Hey & R. W. Allen, Eds.). Anniversary edition DOI: [10.1201/9781003358817](https://doi.org/10.1201/9781003358817); 1996 Addison-Wesley ISBN 0-201-48991-0.

Formula / mechanism systems (Tier A)

1. Udrescu, S.-M., & Tegmark, M. AI Feynman. *Science Advances*. DOI: [10.1126/sciadv.aay2631](https://doi.org/10.1126/sciadv.aay2631).

2. Brunton, S. L., Proctor, J. L., & Kutz, J. N. SINDy. *PNAS*. DOI: [10.1073/pnas.1517384113](https://doi.org/10.1073/pnas.1517384113).
3. Cranmer, M. Interpretable Machine Learning for Science with PySR. arXiv:[2305.01582](https://arxiv.org/abs/2305.01582).
Journal DOI TBD if distinct from arXiv.
4. Fang, Y.-L., Jian, D.-S., Li, X., & Ma, Y.-Q. AI-Newton. arXiv:[2504.01538](https://arxiv.org/abs/2504.01538).
5. Posner, I., Lei, A., & Schölkopf, B. Mechanistic World Models. arXiv:[2607.12474](https://arxiv.org/abs/2607.12474).
6. DeepMind AlphaProof / AlphaGeometry formal-math RL. *Nature* (2025). DOI: [10.1038/s41586-025-09833-y](https://doi.org/10.1038/s41586-025-09833-y).
7. AutoSINDy (hunt note). arXiv:[2605.09696](https://arxiv.org/abs/2605.09696).

Latent / EBM hybrid / spatial world models / SOTA at the plateau (§8.4-§8.7; continued in §8.8-§8.13)

1. Bodnia, E., & Hanin, B. (21 Jan 2026). Energy-Based Models for Reasoning, LLMs for the Interface: Scaling Reasoning with Agentic AI. Logical Intelligence blog. <https://logicalintelligence.com/blog/energy-based-models-for-reasoning>
2. Logical Intelligence. Kona 1.0 product page. <https://logicalintelligence.com/kona-ebms-energy-based-models> (accessed 2026-10-01).
3. Logical Intelligence. Company site (Aleph / leadership / latent reasoning positioning). <https://logicalintelligence.com/> (accessed 2026-10-01).
4. Logical Intelligence. What Sudoku Reveals About AI Reasoning Architectures (Kona demo writeup). <https://logicalintelligence.com/blog/energy-based-model-sudoku-demo> (accessed 2026-10-01). Author-reported Sudoku latency/accuracy figures are field results, not soft-ref measured_j.

Embodiment-class limits (§8.5.1)

1. TypeSafe AI (15 Sep 2026). Introducing System One Models and Jev. <https://typesafe.ai/blog/introducing-system-one-models-and-jev> (accessed 2026-10-04).
2. Cloudflare. Clef model card. <https://huggingface.co/Cloudflare/clef> (accessed 2026-10-04).
3. Liquid AI. Decision Models. <https://docs.liquid.ai/lfm/models/decision-models> (accessed 2026-10-04).
4. Upstage. Solar Decide, Structured Decision API. <https://console.upstage.ai/api/systemone> (accessed 2026-10-04).
5. Perplexity. pplx-decider-v1-27b model card. <https://huggingface.co/perplexity-ai/pplx-decider-v1-27b> (accessed 2026-10-04).
6. Palmer, J. Kev-4B model card. <https://huggingface.co/jaredpalmer/kev-4b> (accessed 2026-10-04).
7. Laya model card. <https://huggingface.co/convaiinnovations/laya> (accessed 2026-10-04).
8. Bespoke Labs. Bespoke-Nimble-9B model card. <https://huggingface.co/bespokelabs/Bespoke-Nimble-9B> (accessed 2026-10-04).

9. iApp Technology. OpenThai-SystemOne model card. <https://huggingface.co/iapp/OpenThai-SystemOne> (accessed 2026-10-04).
10. Decider-4B model card. <https://huggingface.co/Mapika/decider-4b> (accessed 2026-10-04).
11. OpenJev model card. <https://huggingface.co/openjev/openjev> (accessed 2026-10-04).
12. Mica v0.1 4B model card. <https://huggingface.co/sky7350/Mica-v0.1-4B> (accessed 2026-10-04).
13. Winnow-12B model card. <https://huggingface.co/EldanRing/Winnow-12B> (accessed 2026-10-04).
14. Contrastive-LM. CLM-v0.1-8B model card. <https://huggingface.co/Contrastive-LM/CLM-v0.1-8B> (accessed 2026-10-04).
15. Supersonic Labs. Julia 1 model card. <https://huggingface.co/SupersonicLabs/Julia-1> (accessed 2026-10-04).
16. NanoJev model card. <https://huggingface.co/C-Tianyu/NanoJev> (accessed 2026-10-04).
17. Kotoba. open-jev-deberta-v3-large model card. <https://huggingface.co/com-kotobalabs/open-jev-deberta-v3-large> (accessed 2026-10-04).
18. fastText. Text classification tutorial. <https://fasttext.cc/docs/en/supervised-tutorial.html> (accessed 2026-10-04).
19. Google Cloud. Classifying content. <https://docs.cloud.google.com/natural-language/docs/classifying-text> (accessed 2026-10-04).
20. ThinkThen. How answers work. <https://thinkthen.dev/reference/answers/> (accessed 2026-10-04).
21. Fastino. GLiNER2.5-Decide model card. <https://huggingface.co/fastino/GLiNER2.5-Decide> (accessed 2026-10-04).
22. Stately. XState. <https://github.com/statelyai/xstate> (accessed 2026-10-04).
23. Qt Group. Qt State Machine. <https://doc.qt.io/qt-6/qtstatemachine-index.html> (accessed 2026-10-04).
24. Qt Group. Qt SCXML Overview. <https://doc.qt.io/qt-6/qtscxml-overview.html> (accessed 2026-10-04).
25. Boost.Statechart. <https://www.boost.org/doc/libs/latest/libs/statechart/doc/index.html> (accessed 2026-10-04).
26. Boost.MetaStateMachine (Boost 1.86). https://www.boost.org/doc/libs/1_86_0/libs/msm/doc/HTML/index.html (accessed 2026-10-04).

27. boost-ext. SML. <https://github.com/boost-ext/sml> (accessed 2026-10-04).
28. Akka. Classic FSM. <https://doc.akka.io/libraries/akka-core/current/fsm.html> (accessed 2026-10-04).
29. Spring Statemachine. <https://github.com/spring-attic/spring-state-machine> (accessed 2026-10-04).
30. Apache Commons SCXML. <https://commons.apache.org/proper/commons-scxml/index.html> (accessed 2026-10-04).
31. pytransitions. transitions. <https://github.com/pytransitions/transitions> (accessed 2026-10-04).
32. Sismic user manual. <https://sismic.readthedocs.io/en/latest/> (accessed 2026-10-04).
33. dotnet-state-machine. Stateless. <https://github.com/dotnet-state-machine/stateless> (accessed 2026-10-04).
34. Quantum Leaps. QP/C. <https://www.state-machine.com/qpc/> (accessed 2026-10-04).
35. AASM. <https://github.com/aasm/aasm> (accessed 2026-10-04).
36. SCION. <https://github.com/jbeard4/SCION> (accessed 2026-10-04).
37. javascript-state-machine. <https://github.com/jakesgordon/javascript-state-machine> (accessed 2026-10-04).
38. state_machines. https://github.com/state-machines/state_machines (accessed 2026-10-04).
39. HFSM2. <https://github.com/andrew-gresyk/HFSM2> (accessed 2026-10-04).
40. uSCXML. <https://github.com/tklab-tud/uscxml> (accessed 2026-10-04).
41. Stately. <https://stately.ai/> (accessed 2026-10-04).
42. itemis. itemis CREATE. <https://www.itemis.com/en/products/itemis-create/> (accessed 2026-10-04).
43. Quantum Leaps. QM. <https://www.state-machine.com/products/qm> (accessed 2026-10-04).
44. IBM. Engineering Rhapsody. <https://www.ibm.com/products/engineering-rhapsody> (accessed 2026-10-04).
45. Sparx Systems. State Machine Diagram, UML 2 Tutorial. <https://sparxsystems.com/resources/tutorials/uml2/state-diagram.html> (accessed 2026-10-04).
46. Eclipse Papyrus. <https://eclipse.dev/papyrus/> (accessed 2026-10-04).
47. StarUML. <https://staruml.io/> (accessed 2026-10-04).

48. Visual Paradigm. What is State Machine Diagram? <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-state-machine-diagram/> (accessed 2026-10-04).
49. IAR. Embedded development tools (IAR Visual State). <https://www.iar.com/embedded-development-tools> (accessed 2026-10-04).
50. sketch.systems. <https://sketch.systems/> (accessed 2026-10-04).
51. Qt Group. Editing SCXML state charts in Qt Creator. <https://doc.qt.io/qtcreator/creator-scxml.html> (accessed 2026-10-04).
52. Bodnia, E., & Hanin, B. (21 Jan 2026). Energy-Based Models for Reasoning, LLMs for the Interface. Logical Intelligence blog. <https://logicalintelligence.com/blog/energy-based-models-for-reasoning> (accessed 2026-10-04).
53. Toshiba. Simulated Bifurcation Machine, introduction. <https://www.global.toshiba/ww/products-solutions/ai-iot/sbm/intro.html> (accessed 2026-10-04).
54. Fixstars. Amplify annealing machines. <https://amplify.fixstars.com/en/engine> (accessed 2026-10-04).
55. Hitachi (6 Jun 2024). CMOS annealing technology “relaxed MA”. <https://www.hitachi.com/en/press/articles/2024/06/0606/> (accessed 2026-10-04).
56. World Labs. About. <https://www.worldlabs.ai/about> (accessed 2026-10-01).
57. World Labs Team (1 Sep 2026). Atlas: A World Model for Spatial Intelligence. World Labs blog. <https://www.worldlabs.ai/blog/atlas>
58. Li, F.-F. From Words to Worlds: Spatial Intelligence is AI’s Next Frontier (public essay / a16z feature excerpt). <https://www.a16z.news/p/from-words-to-worlds-spatial-intelligence> (accessed 2026-10-01).

Test-time compute / process reward / reasoning models (§8.8)

1. OpenAI (2024). Learning to reason with LLMs (o1 public writeup). <https://openai.com/index/learning-to-reason-with-llms/>
2. Lightman, H., et al. (2023). Let’s Verify Step by Step. arXiv:2305.20050. ICLR 2024.
3. Snell, C., et al. / related inference-time scaling surveys. Inference-Time Scaling for Complex Tasks (survey snapshot). arXiv:2504.00294.
4. Wang, Y., et al. / ACL 2025 line on o1-like test-time scaling. Revisiting the Test-Time Scaling of o1-like Models. ACL 2025. <https://aclanthology.org/2025.acl-long.232.pdf>

Super learning (§8.9)

1. van der Laan, M. J., Polley, E. C., & Hubbard, A. E. (2007). Super Learner. *Statistical Applications in Genetics and Molecular Biology*, 6(1). Related working paper: Polley & van der Laan, Super Learner In Prediction, U.C. Berkeley Biostatistics Working Paper 266. <https://biostats.bepress.com/ucbbiostat/paper266/>
2. Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.
3. Breiman, L. (1996). Stacked regressions. *Machine Learning*, 24, 49–64.

Hardware-paired navigation peers (§8.12.1)

1. Hooker, S. (2021). The hardware lottery. *Communications of the ACM*. doi:10.1145/3467017; arXiv:2009.06489.
2. Huang, W.-H., Sharda, J., Shih, C.-J., et al. (2025). A3D-MoE: Acceleration of Large Language Models with Mixture of Experts via 3D Heterogeneous Integration. arXiv:2507.19142.
3. Chung, J.-W., Wu, R., Ma, J. J., & Chowdhury, M. (2026). Where Do the Joules Go? Diagnosing Inference Energy Consumption. arXiv:2601.22076.

Live SOTA methods for plateau spend (§8.10)

1. Leviathan, Y., Kalman, M., & Matias, Y. (2023). Fast Inference from Transformers via Speculative Decoding. *ICML* (PMLR). <https://proceedings.mlr.press/v202/leviathan23a.html>
2. Shazeer, N., et al. (2017). Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer. *ICLR*. arXiv:1701.06538.
3. Lepikhin, D., et al. (2020). GShard: Scaling Giant Models with Conditional Computation and Automatic Sharding. arXiv:2006.16668.
4. Fedus, W., Zoph, B., & Shazeer, N. (2022). Switch Transformers. *JMLR*, 23(120).
5. Gu, A., & Dao, T. (2023). Mamba: Linear-Time Sequence Modeling with Selective State Spaces. arXiv:2312.00752.
6. Gu, A., Goel, K., & Ré, C. (2022). Efficiently Modeling Long Sequences with Structured State Spaces (S4). *ICLR*.
7. Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*. arXiv:2005.11401.
8. Lou, A., Meng, C., & Ermon, S. (2024). Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution (SEDD). *ICML* (PMLR v235). <https://proceedings.mlr.press/v235/lou24a.html>
9. Sun, Y., et al. (2023). Retentive Network: A Successor to Transformer for Large Language Models. arXiv:2307.08621.

SSM hybrids / MoE / implementation efficiency on the plateau (§8.11)

1. Lieber, O., et al. (2024). Jamba: A Hybrid Transformer-Mamba Language Model. arXiv:2403.19887. AI21 announcement: <https://www.ai21.com/blog/announcing-jamba/>
2. Team Jamba / AI21 (2024). Jamba-1.5: Hybrid Transformer-Mamba Models at Scale. arXiv:2408.12570.

3. Dai, D., et al. (2024). DeepSeekMoE: Towards Ultimate Expert Specialization in Mixture-of-Experts Language Models. *ACL*. <https://aclanthology.org/2024.acl-long.70.pdf>
4. DeepSeek-AI (2024). DeepSeek-V3 Technical Report. arXiv:2412.19437.
5. Guo, D., Yang, D., Zhang, H., et al. / DeepSeek-AI (2025). DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. arXiv:2501.12948. Nature version: Guo et al., *Nature* 645, 633–638 (2025). DOI: [10.1038/s41586-025-09422-z](https://doi.org/10.1038/s41586-025-09422-z).
6. GLM-4.5 Team / Zhipu AI (2025). GLM-4.5: Agentic, Reasoning, and Coding (ARC) Foundation Models. arXiv:2508.06471.
7. GLM Team (2024). ChatGLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools. arXiv:2406.12793.

Plateau techniques: transmission / gearing / cascade / memory-context / test-time (§8.13)

1. Sun, Y., Dong, L., Zhu, Y., et al. (2024). You Only Cache Once: Decoder-Decoder Architectures for Language Models (YOCO). *NeurIPS*. arXiv:2405.05254.
2. DeepSeek-AI (2026). DeepSeek-V4.1-Flash: Pushing the Limits of KV Cache Compression. arXiv:2609.19969. Model card: <https://huggingface.co/deepseek-ai/DeepSeek-V4.1-Flash>. Pricing (field): https://api-docs.deepseek.com/quick_start/pricing.
3. Wei, J., Gao, Y., Zhang, Q., et al. / Xiaomi LLM-Core (2026). HySparse2: Hybrid Sparse Attention with Two-Level KV Sharing. arXiv:2609.26368.
4. Xiaomi LLM-Core (2025). MiMo: Unlocking the Reasoning Potential of Language Model – From Pretraining to Posttraining. arXiv:2505.07608.
5. Xiaomi Team / LLM-Core (2026). MiMo-V2-Flash Technical Report. arXiv:2601.02780.
6. Qwen Team (2025). Qwen3 Technical Report. arXiv:2505.09388.
7. MoonshotAI / Team Kimi (2025). Kimi K2: Open Agentic Intelligence. arXiv:2507.20534.
8. Yang, A., et al. (2023). Baichuan 2: Open Large-scale Language Models. arXiv:2309.10305.
9. Ren, X., et al. (2023). PanGu-Σ: Towards Trillion Parameter Language Model with Sparse Heterogeneous Computing. arXiv:2303.10845.
10. Li, J., et al. (2024). LocMoE: A Low-Overhead MoE for Large Language Model Training. *IJCAI*. arXiv:2401.13920.
11. Sarvam AI (2025–2026). Sarvam-30B MoE model card / docs (≈30B total / ≈2.4B active; 128 experts + shared; top-6). <https://huggingface.co/sarvamai/sarvam-30b> ; <https://docs.sarvam.ai/>
12. Kallappa, A., et al. / Krutrim Team (2025). Krutrim LLM: Multilingual Foundational Model for over a Billion People. arXiv:2502.09642.
13. SB Intuitions (2024). Sarashina2-8x70B MoE release notes / tech blog (8 experts, top-2). <https://www.sbintuitions.co.jp/>
14. Upstage (2026). Solar Open Technical Report. arXiv:2601.07022. Solar Open 2 product materials: <https://www.upstage.ai/>
15. NAVER Cloud / HyperCLOVA (2025). HyperCLOVA X THINK. arXiv:2506.22403.
16. Field note (not a meter): Hacker News thread on DeepSeek-V4.1-Flash KV / prefix-cache economics, item 49735410. <https://news.ycombinator.com/item?id=49735410>

Plateau levers added in industry gap pass (quantization / engines / MoD / edge / neuromorphic / refuse / tools / formal)

1. Frantar, E., Ashkboos, S., Hoefler, T., & Alistarh, D. (2022). GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers. *arXiv*: 2210.17323.
2. Lin, J., Tang, J., Tang, H., et al. (2024). AWQ: Activation-aware Weight Quantization for On-Device LLM Compression and Acceleration. *MLSys*. *arXiv*: 2306.00978.
3. Kwon, W., Li, Z., Zhuang, S., et al. (2023). Efficient Memory Management for Large Language Model Serving with PagedAttention (vLLM). *arXiv*: 2309.06180.
4. Zheng, L., Yin, L., Xie, Z., et al. (2023/2024). SGLang: Efficient Execution of Structured Language Model Programs. *arXiv*: 2312.07104.
5. Raposo, D., Ritter, S., Richards, B., et al. (2024). Mixture-of-Depths: Dynamically allocating compute in transformer-based language models. *arXiv*: 2404.02258.
6. Li, Y., Wei, F., Zhang, C., & Zhang, H. (2024). EAGLE: Speculative Sampling Requires Rethinking Feature Uncertainty. *ICML (PMLR)*. *arXiv*: 2401.15077.
7. Bai, Y., Kadavath, S., Kundu, S., et al. (2022). Constitutional AI: Harmlessness from AI Feedback. *arXiv*: 2212.08073.
8. Schick, T., Dwivedi-Yu, J., Dessì, R., et al. (2023). Toolformer: Language Models Can Teach Themselves to Use Tools. *NeurIPS*. *arXiv*: 2302.04761.
9. Yao, S., Zhao, J., Yu, D., et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. *ICLR*. *arXiv*: 2210.03629.
10. Yang, K., Swope, A., Gu, A., et al. (2023). LeanDojo: Theorem Proving with Retrieval-Augmented Language Models. *NeurIPS*. *arXiv*: 2306.15626.
11. Lin, J., Chen, W.-M., Lin, Y., et al. (2020). MCUNet: Tiny Deep Learning on IoT Devices. *NeurIPS*. *arXiv*: 2007.10319.
12. Wei, J., et al. (2024). T-MAC: CPU Renaissance via Table Lookup for Low-Bit LLM Deployment on Edge. *arXiv*: 2407.00088.
13. Abreu, S., Shrestha, S. B., Zhu, R.-J., & Eshraghian, J. (2025). Neuromorphic Principles for Efficient Large Language Models on Intel Loihi 2. *arXiv*: 2503.18002.
14. CN101 digital thermodynamic computer for generative AI. *arXiv*: 2608.00754 (field / soft-ref only; not soft-ref measured_j).
15. Zilberstein, S. (1996). Using Anytime Algorithms in Intelligent Systems. *AI Magazine*, 17(3).
16. Tishby, N., Pereira, F. C., & Bialek, W. (1999). The information bottleneck method. *Allerton / arXiv physics:0004057*. Tishby & Zaslavsky (2015), Deep Learning and the Information Bottleneck Principle, *arXiv*: 1503.02406.

Race to plateau floors across AI/ML fields (§8.14)

1. Ouyang, L., Wu, J., Jiang, X., et al. (2022). Training language models to follow instructions with human feedback (InstructGPT / RLHF). *NeurIPS*. *arXiv*:[2203.02155](#).
2. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-Resolution Image Synthesis with Latent Diffusion Models. *CVPR*. *arXiv*:[2112.10752](#).
3. Salimans, T., & Ho, J. (2022). Progressive Distillation for Fast Sampling of Diffusion Models. *ICLR*. *arXiv*:[2202.00512](#).

4. Blattmann, A., et al. (2023). Stable Video Diffusion: Scaling Latent Video Diffusion Models to Large Datasets. arXiv:[2311.15127](#).
5. Kipf, T. N., & Welling, M. (2017). Semi-Supervised Classification with Graph Convolutional Networks. *ICLR*. arXiv:[1609.02907](#).
6. Radford, A., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision (CLIP). *ICML* (PMLR v139).
7. McMahan, H. B., Moore, E., Ramage, D., Hampson, S., & Agüera y Arcas, B. (2017). Communication-Efficient Learning of Deep Networks from Decentralized Data (FedAvg). *AISTATS* (PMLR v54).
8. Kirkpatrick, J., et al. (2017). Overcoming catastrophic forgetting in neural networks (EWC). *PNAS*. DOI: [10.1073/pnas.1611835114](#).
9. Frankle, J., & Carbin, M. (2019). The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks. *ICLR*. arXiv:[1803.03635](#).
10. Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the Knowledge in a Neural Network. arXiv:[1503.02531](#).
11. Zoph, B., & Le, Q. V. (2017). Neural Architecture Search with Reinforcement Learning. *ICLR*. arXiv:[1611.01578](#).
12. Cai, H., Gan, C., Wang, T., Zhang, Z., & Han, S. (2020). Once-for-All: Train One Network and Specialize it for Efficient Deployment. *ICLR*. arXiv:[1908.09791](#).
13. Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. *ICML* (PMLR v48). arXiv:[1506.02142](#).
14. Angelopoulos, A. N., & Bates, S. (2021/2023). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv:[2107.07511](#); *Foundations and Trends in Machine Learning*.
15. Vovk, V., Gammerman, A., & Shafer, G. (2022). *Algorithmic Learning in a Random World* (2nd ed.). Springer. DOI: [10.1007/978-3-031-06649-8](#).
16. Pearl, J. (2009). Causal inference in statistics: An overview. *Statistics Surveys*, 3, 96–146. See also Pearl, *Causality* (2nd ed.).
17. Settles, B. (2009). Active Learning Literature Survey. University of Wisconsin–Madison Computer Sciences Technical Report 1648.
18. NVIDIA. CUDA Programming Guide — CUDA Graphs (documentation). <https://docs.nvidia.com/cuda/cuda-programming-guide/> (accessed 2026-10-02). PyTorch CUDA Graphs / torch.compile reduce-overhead notes: <https://pytorch.org/docs/stable/notes/cuda.html>
19. Packer, C., et al. (2023). MemGPT: Towards LLMs as Operating Systems. arXiv:[2310.08560](#).
20. Edge, D., et al. (2024). From Local to Global: A GraphRAG Approach to Query-Focused Summarization. arXiv:[2404.16130](#).
21. Willard, B. T., & Louf, R. (2023). Efficient Guided Generation for Large Language Models (Outlines). arXiv:[2307.09702](#).

Claim regimes, labels, and facility duties (§11.4.1)

1. Ministère de la Transition écologique et de la Cohésion des territoires. Publication du référentiel général pour l'IA frugale. Press release, 28 June 2024, updated 20 July 2026. <https://www.ecologie.gouv.fr/presse/publication-du-referentiel-general-lia-frugale-sattaquer-limpact-environnemental-lia>
2. AFNOR. Un référentiel pour mesurer et réduire l'impact environnemental de l'IA. 12 July 2024, updated 17 December 2025. <https://www.afnor.org/actualites/intelligence-artificielle/referentiel-reduire-impact-environnemental-ia/>
3. AFNOR. AFNOR SPEC 2314, *Référentiel général pour l'IA frugale*. June 2024. 100 pages. Shop record: <https://www.boutique.afnor.org/en-gb/standard/afnor-spec-2314/fa208976/421140>
4. Umweltbundesamt (2020). Umweltzeichen Blauer Engel, Ressourcen- und energieeffiziente Softwareprodukte (DE-UZ 215). Ausgabe Januar 2020, Version 6 (July 2026). RAL gGmbH (Hrsg.). <https://produktinfo.blauer-engel.de/uploads/criteriafile/de/171/DE-UZ%20215-202001-de%20Kriterien-V6.pdf> Product page: <https://www.blauer-engel.de/de/produktwelt/software>
5. Schulze, M., Gröger, J., and Behrens, F. SoftAWERE. Umweltbundesamt TEXTE 71/2026. Completion date May 2023. <https://www.umweltbundesamt.de/en/publikationen/softawere> PDF: https://www.umweltbundesamt.de/system/files/medien/11850/publikationen/2026-05/71_2026_TEXTE.pdf
6. UNE. ESPECIFICACION UNE 0086:2025, *Medición del consumo energético, huella de carbono, consumo del agua y rendimiento de sistemas de Inteligencia Artificial*. Edition 8 October 2025, corrected 29 July 2026. <https://www.une.org/encuentra-tu-norma/busca-tu-norma/norma?c=N0075012>
7. UNE. UNE promotes the first specification to measure the sustainability of Artificial Intelligence. 10 October 2025. <https://www.en.une.org/la-asociacion/sala-de-informacion-une/noticias/nueva-especificacion-une-para-medir-la-sostenibilidad-de-la-ia>
8. Secretaría de Estado de Digitalización e Inteligencia Artificial. España crea una pionera especificación UNE que medirá la sostenibilidad de la Inteligencia Artificial. 9 October 2025. <https://digital.gob.es/comunicacion/notas-prensa/secretaria-digitalizacion-e-inteligencia-artificial/2025/10/2025-10-09>
9. 市场监管总局 and 国家发展改革委. 《人工智能计量体系和能力建设指引（2026版）》. Notice, 28 May 2026. https://www.samr.gov.cn/xw/zj/art/2026/art_f43aa2c974654d66b91bbad8410d0d71.html
10. BSI. PD CEN/CLC/TR 18145:2025, *Environmentally sustainable Artificial Intelligence*. Published 28 February 2025. <https://knowledge.bsigroup.com/products/environmentally-sustainable-artificial-intelligence>
11. CEN-CENELEC. Gap Analysis of Standards for EU AI Act Compliance for healthcare Artificial Intelligence, Task 2 Mid-Phase Report. 18 December 2025. Lists CEN/CLC/TR 18145:2025 (WI JT021010) as published in the JTC 21 column. https://www.cencenelec.eu/media/CEN-CENELEC/AreasOfWork/CEN%20sectors/Healthcare/cen-clc_sag_healthcarestandards-midphasereportartificialintelligence.pdf

12. CEN and CENELEC. Update on CEN and CENELEC's Decision to Accelerate the Development of Standards for Artificial Intelligence. Posted 23 October 2025. <https://www.cencenelec.eu/news-events/news/2025/brief-news/2025-10-23-ai-standardization/>
13. 資源エネルギー庁. 増加が見込まれるデータセンターの電力需要をどうする？さらなる省エネを進める新たな制度に注目！ Explainer, 20 May 2026, correction noted 4 June 2026. https://www.enecho.meti.go.jp/about/special/johoteikyo/data_center2026.html

Energy as the condition (§11.4.2)

1. Margolus, N., & Levitin, L. B. (1998). The maximum speed of dynamical evolution. *Physica D*, 120, 188-195. DOI: [10.1016/S0167-2789\(98\)00054-2](https://doi.org/10.1016/S0167-2789(98)00054-2). arXiv:[quant-ph/9710043](https://arxiv.org/abs/quant-ph/9710043).
2. Sagawa, T., & Ueda, M. (2009). Minimal energy cost for thermodynamic information processing: measurement and information erasure. *Physical Review Letters*, 102, 250602. DOI: [10.1103/PhysRevLett.102.250602](https://doi.org/10.1103/PhysRevLett.102.250602). Erratum: *Physical Review Letters*, 106, 189901 (2011).
3. Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics of prediction. *Physical Review Letters*, 109, 120604. DOI: [10.1103/PhysRevLett.109.120604](https://doi.org/10.1103/PhysRevLett.109.120604).
4. Bérut, A., Arakelyan, A., Petrosyan, A., Ciliberto, S., Dillenschneider, R., & Lutz, E. (2012). Experimental verification of Landauer's principle linking information and thermodynamics. *Nature*, 483, 187-189. DOI: [10.1038/nature10872](https://doi.org/10.1038/nature10872).
5. Jun, Y., Gavrilov, M., & Bechhoefer, J. (2014). High-precision test of Landauer's principle in a feedback trap. *Physical Review Letters*, 113, 190601. DOI: [10.1103/PhysRevLett.113.190601](https://doi.org/10.1103/PhysRevLett.113.190601).
6. Kolchinsky, A., & Wolpert, D. H. (2020). Thermodynamic costs of Turing machines. *Physical Review Research*, 2, 033312. DOI: [10.1103/PhysRevResearch.2.033312](https://doi.org/10.1103/PhysRevResearch.2.033312).
7. You, J., Chung, J.-W., & Chowdhury, M. (2023). Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training. *20th USENIX Symposium on Networked Systems Design and Implementation (NSDI 23)*, 119-139. <https://www.usenix.org/conference/nsdi23/presentation/you>. arXiv:[2208.06102](https://arxiv.org/abs/2208.06102).
8. Chung, J.-W., Gu, Y., Jang, I., Meng, L., Bansal, N., & Chowdhury, M. (2024). Reducing Energy Bloat in Large Model Training. *ACM SIGOPS 30th Symposium on Operating Systems Principles (SOSP '24)*. DOI: [10.1145/3694715.3695970](https://doi.org/10.1145/3694715.3695970). arXiv:[2312.06902](https://arxiv.org/abs/2312.06902).

Neuromorphic substrates and measured-run energy (§11.4.3)

1. Merolla, P. A., Arthur, J. V., Alvarez-Icaza, R., Cassidy, A. S., Sawada, J., Akopyan, F., Jackson, B. L., Imam, N., Guo, C., Nakamura, Y., Brezzo, B., Vo, I., Esser, S. K., Appuswamy, R., Taba, B., Amir, A., Flickner, M. D., Risk, W. P., Manohar, R., & Modha, D. S. (2014). A million spiking-neuron integrated circuit with a scalable communication network and interface. *Science*, 345(6197), 668-673. DOI: [10.1126/science.1254642](https://doi.org/10.1126/science.1254642).
2. Frenkel, C., Lefebvre, M., Legat, J.-D., & Bol, D. (2018). A 0.086-mm² 12.7-pJ/SOP 64k-Synapse 256-Neuron Online-Learning Digital Spiking Neuromorphic Processor in 28nm CMOS. *IEEE Transactions on Biomedical Circuits and Systems*. DOI: [10.1109/TBCAS.2018.2880425](https://doi.org/10.1109/TBCAS.2018.2880425). arXiv:[1804.07858](https://arxiv.org/abs/1804.07858).

3. Esser, S. K., Merolla, P. A., Arthur, J. V., Cassidy, A. S., Appuswamy, R., Andreopoulos, A., Berg, D. J., McKinstry, J. L., Melano, T., Barch, D. R., di Nolfo, C., Datta, P., Amir, A., Taba, B., Flickner, M. D., & Modha, D. S. (2016). Convolutional networks for fast, energy-efficient neuromorphic computing. *Proceedings of the National Academy of Sciences*, 113(41), 11441-11446. DOI: [10.1073/pnas.1604850113](https://doi.org/10.1073/pnas.1604850113). arXiv:[1603.08270](https://arxiv.org/abs/1603.08270).
4. Göltz, J., Kriener, L., Baumbach, A., Billaudelle, S., Breitwieser, O., Cramer, B., Dold, D., Kungl, A. F., Senn, W., Schemmel, J., Meier, K., & Petrovici, M. A. (2021). Fast and energy-efficient neuromorphic deep learning with first-spike times. *Nature Machine Intelligence*, 3, 823-835. DOI: [10.1038/s42256-021-00388-x](https://doi.org/10.1038/s42256-021-00388-x). arXiv:[1912.11443](https://arxiv.org/abs/1912.11443).
5. Blouw, P., Choo, X., Hunsberger, E., & Elias Smith, C. (2019). Benchmarking Keyword Spotting Efficiency on Neuromorphic Hardware. *Proceedings of the 7th Annual Neuro-inspired Computational Elements Workshop (NICE '19)*. DOI: [10.1145/3320288.3320304](https://doi.org/10.1145/3320288.3320304). arXiv:[1812.01739](https://arxiv.org/abs/1812.01739).
6. Gobieski, G., Lucia, B., & Beckmann, N. (2019). Intelligence Beyond the Edge: Inference on Intermittent Embedded Systems. *Proceedings of the Twenty-Fourth International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '19)*, 199-213. DOI: [10.1145/3297858.3304011](https://doi.org/10.1145/3297858.3304011). arXiv:[1810.07751](https://arxiv.org/abs/1810.07751).
7. Samsi, S., Zhao, D., McDonald, J., Li, B., Michaleas, A., Jones, M., Bergeron, W., Kepner, J., Tiwari, D., & Gadepally, V. (2023). From Words to Watts: Benchmarking the Energy Costs of Large Language Model Inference. *2023 IEEE High Performance Extreme Computing Conference (HPEC)*. DOI: [10.1109/HPEC58863.2023.10363447](https://doi.org/10.1109/HPEC58863.2023.10363447). arXiv:[2310.03003](https://arxiv.org/abs/2310.03003).
8. Luccioni, A. S., Jernite, Y., & Strubell, E. (2024). Power Hungry Processing: Watts Driving the Cost of AI Deployment? *ACM Conference on Fairness, Accountability, and Transparency (FAccT '24)*, 85-99. DOI: [10.1145/3630106.3658542](https://doi.org/10.1145/3630106.3658542). arXiv:[2311.16863](https://arxiv.org/abs/2311.16863).
9. Yang, H., Zhu, Y., & Liu, J. (2019). Energy-Constrained Compression for Deep Neural Networks via Weighted Sparse Projection and Layer Input Masking. *International Conference on Learning Representations (ICLR)*. arXiv:[1806.04321](https://arxiv.org/abs/1806.04321).

Analog, in-memory, and frequency-domain computing (§11.4.4)

1. Ambrogio, S., Narayanan, P., Tsai, H., Shelby, R. M., Boybat, I., di Nolfo, C., Sidler, S., Giordano, M., Bodini, M., Farinha, N. C. P., Killeen, B., Cheng, C., Jaoudi, Y., & Burr, G. W. (2018). Equivalent-accuracy accelerated neural-network training using analogue memory. *Nature*, 558, 60-67. DOI: [10.1038/s41586-018-0180-5](https://doi.org/10.1038/s41586-018-0180-5).
2. Ambrogio, S., Narayanan, P., Okazaki, A., Fasoli, A., Mackin, C., Hosokawa, K., Nomura, A., Yasuda, T., Chen, A., Friz, A., Ishii, M., Luquin, J., Kohda, Y., Saulnier, N., Brew, K., Choi, S., Ok, I., Philip, T., Chan, V., Silvestre, C., Ahsan, I., Narayanan, V., Tsai, H., & Burr, G. W. (2023). An analog-AI chip for energy-efficient speech recognition and transcription. *Nature*, 620, 768-775. DOI: [10.1038/s41586-023-06337-5](https://doi.org/10.1038/s41586-023-06337-5).
3. Le Gallo, M., Khaddam-Aljameh, R., Stanisavljevic, M., Vasilopoulos, A., Kersting, B., Dazzi, M., Karunaratne, G., Brändli, M., Singh, A., Müller, S. M., Büchel, J., Timoneda, X., Joshi, V., Rasch, M. J., Egger, U., Garofalo, A., Petropoulos, A., Antonakopoulos, T., Brew, K., Choi, S., Ok, I., Philip, T., Chan, V., Silvestre, C., Ahsan, I., Saulnier, N., Narayanan, V., Francese, P. A.,

- Eleftheriou, E., & Sebastian, A. (2023). A 64-core mixed-signal in-memory compute chip based on phase-change memory for deep neural network inference. *Nature Electronics*, 6, 680-693. DOI: [10.1038/s41928-023-01010-1](https://doi.org/10.1038/s41928-023-01010-1). arXiv:[2212.02872](https://arxiv.org/abs/2212.02872).
4. Wan, W., Kubendran, R., Schaefer, C., Eryilmaz, S. B., Zhang, W., Wu, D., Deiss, S., Raina, P., Qian, H., Gao, B., Joshi, S., Wu, H., Wong, H.-S. P., & Cauwenberghs, G. (2022). A compute-in-memory chip based on resistive random-access memory. *Nature*, 608, 504-512. DOI: [10.1038/s41586-022-04992-8](https://doi.org/10.1038/s41586-022-04992-8).
 5. Yao, P., Wu, H., Gao, B., Tang, J., Zhang, Q., Zhang, W., Yang, J. J., & Qian, H. (2020). Fully hardware-implemented memristor convolutional neural network. *Nature*, 577, 641-646. DOI: [10.1038/s41586-020-1942-4](https://doi.org/10.1038/s41586-020-1942-4).
 6. Shen, Y., Harris, N. C., Skirlo, S., Prabhu, M., Baehr-Jones, T., Hochberg, M., Sun, X., Zhao, S., Larochelle, H., Englund, D., & Soljačić, M. (2017). Deep learning with coherent nanophotonic circuits. *Nature Photonics*, 11, 441-446. DOI: [10.1038/nphoton.2017.93](https://doi.org/10.1038/nphoton.2017.93). arXiv:[1610.02365](https://arxiv.org/abs/1610.02365).
 7. Wang, T., Ma, S.-Y., Wright, L. G., Onodera, T., Richard, B. C., & McMahon, P. L. (2022). An optical neural network using less than 1 photon per multiplication. *Nature Communications*, 13, 123. DOI: [10.1038/s41467-021-27774-8](https://doi.org/10.1038/s41467-021-27774-8). arXiv:[2104.13467](https://arxiv.org/abs/2104.13467).
 8. Chen, Y., Nazhamaiti, M., Xu, H., Meng, Y., Zhou, T., Li, G., Fan, J., Wei, Q., Wu, J., Qiao, F., Fang, L., & Dai, Q. (2023). All-analog photoelectronic chip for high-speed vision tasks. *Nature*, 623, 48-57. DOI: [10.1038/s41586-023-06558-8](https://doi.org/10.1038/s41586-023-06558-8).
 9. Feldmann, J., Youngblood, N., Karpov, M., Gehring, H., Li, X., Stappers, M., Le Gallo, M., Fu, X., Lukashchuk, A., Raja, A. S., Liu, J., Wright, C. D., Sebastian, A., Kippenberg, T. J., Pernice, W. H. P., & Bhaskaran, H. (2021). Parallel convolutional processing using an integrated photonic tensor core. *Nature*, 589, 52-58. DOI: [10.1038/s41586-020-03070-1](https://doi.org/10.1038/s41586-020-03070-1).
 10. Miscuglio, M., Hu, Z., Li, S., George, J. K., Capanna, R., Dalir, H., Bardet, P. M., Gupta, P., & Sorger, V. J. (2020). Massively parallel amplitude-only Fourier neural network. *Optica*, 7(12), 1812. DOI: [10.1364/OPTICA.408659](https://doi.org/10.1364/OPTICA.408659). arXiv:[2008.05853](https://arxiv.org/abs/2008.05853).
 11. Moy, W., Ahmed, I., Chiu, P.-W., Moy, J., Sapatnekar, S. S., & Kim, C. H. (2022). A 1,968-node coupled ring oscillator circuit for combinatorial optimization problem solving. *Nature Electronics*, 5, 310-317. DOI: [10.1038/s41928-022-00749-3](https://doi.org/10.1038/s41928-022-00749-3).
 12. Romera, M., Talatchian, P., Tsunegi, S., Abreu Araujo, F., Cros, V., Bortolotti, P., Trastoy, J., Yakushiji, K., Fukushima, A., Kubota, H., Yuasa, S., Ernoult, M., Vodenicarevic, D., Hirtzlin, T., Locatelli, N., Querlioz, D., & Grollier, J. (2018). Vowel recognition with four coupled spin-torque nano-oscillators. *Nature*, 563, 230-234. DOI: [10.1038/s41586-018-0632-y](https://doi.org/10.1038/s41586-018-0632-y). arXiv:[1711.02704](https://arxiv.org/abs/1711.02704).
 13. Torrejon, J., Riou, M., Abreu Araujo, F., Tsunegi, S., Khalsa, G., Querlioz, D., Bortolotti, P., Cros, V., Yakushiji, K., Fukushima, A., Kubota, H., Yuasa, S., Stiles, M. D., & Grollier, J. (2017). Neuromorphic computing with nanoscale spintronic oscillators. *Nature*, 547, 428-431. DOI: [10.1038/nature23011](https://doi.org/10.1038/nature23011). arXiv:[1701.07715](https://arxiv.org/abs/1701.07715).
 14. Zhong, Y., Tang, J., Li, X., Liang, X., Liu, Z., Li, Y., Xi, Y., Yao, P., Hao, Z., Gao, B., Qian, H., & Wu, H. (2022). A memristor-based analogue reservoir computing system for real-time and power-efficient signal processing. *Nature Electronics*, 5, 672-681. DOI: [10.1038/s41928-022-00838-3](https://doi.org/10.1038/s41928-022-00838-3).

Dual-phase perception / TinyML transducers (§11.1)

1. Lin, J., Chen, W.-M., Lin, Y., et al. (2020). MCUNet: Tiny Deep Learning on IoT Devices. *NeurIPS*. arXiv:[2007.10319](https://arxiv.org/abs/2007.10319). (Also listed as ref. 74; retained here for §11.1 locality.)
2. Zhang, Y., Sun, S., & Ma, L. (2021). Tiny Transducer: A Highly-Efficient Speech Recognition Model on Edge Devices. *ICASSP*. arXiv:[2101.06856](https://arxiv.org/abs/2101.06856). DOI: [10.1109/ICASSP39728.2021.9413854](https://doi.org/10.1109/ICASSP39728.2021.9413854).

OpenIE companions

1. Charlot, D. Notational Intelligence as Commit Law. research.openie.dev/papers/ni/.
2. Charlot, D. Satiation and Scarcity after Free AI. research.openie.dev/papers/satiation/.
3. OpenIE Periodic Stack of Computation; compute.openie.dev (258 primitives / 33 families). 50a. OpenIE Stack (family map); stack.openie.dev. 50b. OpenIE Knowledge (seven-axis claim); knowledge.openie.dev. 50c. OpenIE Synthesis (DIS / zones / cost surface); synthesis.openie.dev. 50d. OpenIE Verify (verification stack); proof.openie.dev.

Kepler's laws / Newton's *Principia* / Brahe's observational program are treated as historical common knowledge in §8.1; imprint DOIs for critical editions **TBD** if a journal version requires them. Raiffa decision-analysis imprint DOI **TBD**. Press-wire details for Logical Intelligence beyond the company blog (exact pilot lists, internal AGI claims) are company-reported. Author-reported GPU-hours, throughput, KV bytes/token, prefill FLOPs, API cache prices, and benchmark scores for Jamba, DeepSeek (including V4.1-Flash), GLM, Xiaomi MiMo / HySparse2, Qwen, Kimi, Sarvam, Solar, speculative decoding, and test-time scaling papers are **field / author-reported**, not soft-ref measured_j. Hacker News cache-break anecdotes are operator reports, not OpenIE meters.

Appendix A. Measurement bounds

- Soft-ref prove path: board_synth_claimed=false; measured_j=None unless a Metered probe returns
- Landauer = labeled thermodynamic lower-bound estimate on receipts
- Analytical / catalog estimated_j ≠ board power
- Soft-ref mol prove ~29 VERIFIED = constructive existence in **software**
- Clean-room Mixture of Limits: law+runtime, not an MoE wrapper
- Bottleneck: materials/applied-math embodiment. Settled IT/physics already name the floors
- §8.4-§8.14 map latents / Logical Intelligence / World Labs / **SOTA at the plateau** (test-time / super learning / live SOTA methods including MoD / quantization / inference engines / SSM—MoE—implementation efficiency / hardware economics / edge—neuromorphic soft-ref / transmission—gearing—cascade—memory—context—test-time / **race to plateau floors across AI/ML fields**) as plateau spend options under Mixture of Limits floors; the industry race is to those floors, not unbounded scale; §9 contrasts Constitutional AI and tool-use agents with typed certify-before-commit; field product joules, API prices, and author-reported KV bytes stay field; estimates ≠ board package energy

- §11 closes engineering gaps as plateau work: dual-phase micro-perception (§11.1; rule AST path shipped), O(1) meta-routing (§11.2), Primitive Distillation v1 (§11.3; `mol distill`), Tier 0/1/2 measurement realism (§11.4; only Metered populates `measured_j`; Tier-2 shunt HAL stub honesty), DX `mol.yaml` / `mol prove A1–A14` / `mol bench` / `mol arena` / `mol phase1` / `mol distill` (§11.5)—shipped in clean-room product/; `mol dev` UI polish residual. Soft-ref in proof: Ferric inventory, Stage C inventory, HW Gaps sims x8 (`physical_settle...` `photonic_mzi`; Gap cells retained). Out of soft-ref: Ferric robot EFA meters, live WCA MCP network (HTTPIMCP certify env-gated in-tree), FPGA Stage C package meters (`stage_c_measured=false`)
- Dual-phase does not demote Lookup → Formula → Solver → Model LAST; Phase 1 emits typed AST/schema; Model LAST remains residual leaf in Phase 2

Appendix B. Reproducibility pointers

```
# On a machine with the mixture-of-limits workspace:
cargo test --workspace
cargo run -p mol-cli -- prove # expect VERIFIED lines, exit 0
```

Site preview for this study (research-openie-web):

```
# From the research-openie-web checkout root:
pnpm install
pnpm dev
# open /papers/mol/ , /about/mol , /living/mol/
```

PDF regeneration (optional; needs pandoc + weasyprint): `./scripts/generate-pdfs.sh` after `mol` is added to the script's `gen_one` list.